

Physics-Infused Reduced-Order Modeling for Analysis of Multi-Layered Hypersonic Thermal Protection Systems With Imperfect Thermal Contacts

Abstract

This work presents a systematic design and implementation of the physics-infused reduced-order modeling (PIROM) framework for optimal inference and reduced-order modeling of hypersonic TPS with unknown thermal contact resistances (TCRs). The PIROM framework is designed to leverage the strengths of both physics-based and data-driven modeling paradigms, by infusing physics-based models with data-driven components to enhance their predictive capabilities and generalizability. Inference heat conduction problems (IHCPs) are solved using the PIROM framework to infer TCRs from internal temperature measurements, while providing accurate predictions of transient TPS temperature dynamics under varying boundary conditions and material properties. Particularly, experimental limitations are considered in the design of the PIROM framework, including noisy observables and thermocouple placement limitations. The performance of the PIROM framework is benchmarked against traditional physics-based modeling approaches (e.g., one-dimensional FEM) and data-driven surrogate models (e.g., neural ordinary differential equations) for both inferring contact conductances and transient TPS temperature dynamics.

1 Introduction

What is the importance of hypersonic TPS? Hypersonic thermal protection systems (TPS) play a critical role in safeguarding hypersonic vehicles as they are exposed to extreme aerodynamic heating during flight. Therefore, accurate modeling and prediction of TPS behavior become essential for designing effective thermal protection strategies. Nonetheless, the highly non-linear fluid-thermal-structural interactions, temperature-dependent material property variations, and uncertainties in thermal contact resistances (TCR) at the interfaces of the TPS layers, pose significant challenges for traditional modeling approaches such as finite-element methods (FEM), as well as reduced-order (ROM) and surrogate (SM) modeling techniques.

What is the new problem that we want to solve? Particularly, a critical artifact in transient reduced-order modeling of hypersonic TPS is the presence of TCRs. Due to manufacturing limitations, when two solids are brought together to form an interface, the true solid-to-solid contact area between them is generally a small fraction of the apparent area over which they meet [Minges1970]. The net effect of such imperfect contacts is a temperature discontinuity at the interface that is in general challenging to predict, and that significantly affects the heat transfer across the interface. The identification of TPS properties from internal temperature measurements belongs to a class of *inverse heat conduction problem* (IHCP) [Deng2026], which are known to be ill-posed and sensitive to measurement noise, and thus require careful regularization and robust inference techniques for accurate estimation of TCRs. The challenges posed by IHCPs can introduce significant

uncertainties in transient temperature predictions, and thus compromise the reliability of ROMs for design, analysis, and optimization of hypersonic TPS [Fang2023]. Therefore, there is a critical need for *efficient, accurate, and generalizable* modeling approaches of TPS that can infer TCRs, while providing reliable predictions of internal TPS transient temperatures under varying boundary conditions and material properties.

What has been done in the literature for inference and simulation? Attempts to model hypersonic TPS with varying TCRs have taken various forms in the literature, ranging from traditional deterministic physics-based and probabilistic approaches

For instance, in Ref. [Fang2023], a Lattice Boltzmann Method (LBM) is used to model the transient heat transfer and solid-liquid phase change in a hypersonic TPS, where the TCRs are assumed known, and are shown to significantly delay the melting points of phase changing materials. In Ref. [Zhang2024], authors take one step further by treating the TCRs as unknown parameters, and a finite-element thermo-mechanical model of a three-dimensional reusable heat-pipe cooled thermal protection system is developed to perform Levenberg-Marquardt gradient-based inference of internal TCRs using boundary temperature measurements. A similar approach is adopted in Ref. [Brociek2024], where a finite-difference one-dimensional model of a three-layered TPS is developed to solve the forward problem, and a Levenberg-Marquardt for a Tikhonov regularized optimization is used to solve the inverse problem of inferring TCRs from internal temperature measurements.

Probabilistic approaches to solving the inverse problem of inferring TCRs have also been proposed in the literature. For instance, in Ref. [Gao2026], a Bayesian Network (BN) with sequential importance resample (SIR) is leveraged for probabilistic inversion of TCRs, where the BN is implemented by leveraging a finite-element model of the multi-layered TPS for forward simulations. A BN is also leveraged in Ref. [Deng2026] to infer TCRs, where a neural network is used as a surrogate model to predict internal temperature dynamics due to the computational cost of the FEM-based forward model.

The use of scientific machine learning (SciML) techniques for modeling hypersonic TPS with unknown TCRs has limited exploration in the literature.

.... Other X approaches such as those in Refs.[X] ...

While these methods offer a high-fidelity representation of the underlying physics, highly refined FEMs result in large-scale systems that are computationally expensive to solve, and thus are not suitable for design, analysis, and optimization of hypersonic TPS.

Other data-driven approaches such as those in Refs.[X] ..., leverage Bayesian Networks (BNs) to learn mappings from boundary conditions and thermocop

machine learning techniques to learn the underlying dynamics of the TPS system directly from data, while inferring the TCRs. For instance, in Refs. [X]...

While these methods offer computational efficiency, they often require large amounts of data for training, and may not generalize well to unseen conditions, such as varying boundary conditions and material properties. Moreover, they may not provide insights into the underlying physics of the system, which can be critical for design and optimization.

What is the present work doing? This work presents a systematic design, implementation, and benchmarking of the PIRM for modeling hypersonic TPS with unknown TCRs under experimental constraints such as noisy observables and thermocouple placement limitations.

Objectives of the work The objectives of this work are to,

1. Demonstrate the systematic application of PIROM for modeling hypersonic TPS with unknown contact resistances, emphasizing on the simultaneous inference of material properties and hidden dynamics with limited data.
2. Benchmark the performance of PIROM against traditional physics-based modeling approaches (e.g, one-dimensional FEM) and data-driven surrogate models (e.g., neural ordinary differential equations) for both inferring contact conductances and transient TPS temperature dynamics.
3. Provide insights into the optimal design of physics-based models for inference of contact conductances, including the resolution of the physics-based model, noise levels, and constraints with the placement of thermocouples.

2 Modeling of Hypersonic Thermal Protection Systems

The following discussion presents the modeling of the transient heat conduction problem in multi-layered hypersonic thermal protection systems (TPS) with imperfect interfaces applications. Specifically, three different approaches of decreasing fidelity are presented for solving the governing equations: (i) a high-resolution three-dimensional FEM termed the *full-order model* (FOM), (ii) a one-dimensional through-thickness FEM with linear elements termed the *one-dimensional FEM* (1D-FEM), and (iii) a lumped capacitance model (LCM) representation of the TPS with interfacial temperature jump dynamics termed the *reduced-physics model* (RPM). The three solution strategies are described in detail next.

2.1 Governing Equations

Consider the generic domain $\Omega \subset \mathbb{R}^d$ show in Fig. 1. A heat flux q_b (i.e., Neumann boundary condition) is prescribed on boundary e_q , while a temperature T_b (i.e., Dirichlet boundary condition) is prescribed on boundary e_T , and $\partial\Omega = \cup_{i=1}^N \partial\Omega_i$ where $e_{iq}, e_{iT} \subset \partial\Omega_i$. The transient heat transfer is governed by the following energy PDE,

$$\rho c_p \partial_t T - \nabla \cdot (\mathbf{k}(T) \nabla T) = 0, \quad x \in \Omega \quad (1a)$$

$$-\mathbf{k}(T) \nabla T \cdot \mathbf{n} = q_b(r, t), \quad r \in e_q \quad (1b)$$

$$T(r, t) = T_b(r, t), \quad r \in e_T \quad (1c)$$

$$T(r, 0) = T_0(r), \quad x \in \Omega \quad (1d)$$

where the density ρ is constant, while the heat capacity c_p and thermal conductivity $\mathbf{k} \in \mathbb{R}^{d \times d}$ are temperature-dependent. The TPS domain is further divided into N non-overlapping components $\{\Omega_i\}_{i=1}^N$, with N_Γ different imperfect interfaces between components. Specifically, to facilitate subsequent model derivations, each domain Ω_i is further subdivided into: (i) a *contact subdomain* Ω_i^c consisting of the portion of TPS layer that is in direct contact with a neighboring layer across and imperfect interface, and (ii) a *bulk subdomain* Ω_i^b consisting of the remaining interior bulk material; clearly, $\Omega_i = \Omega_i^b \cup \Omega_i^c$, and $\Omega_i^b \cap \Omega_i^c = \emptyset$.

For example, Fig. 1 contains $N_C = 3$ components and $N_\Gamma = 3$ imperfect contacts between components. Such partition of the TPS is introduced to systematically model the *bulk* and *contact* heat transfer as separate but coupled mechanisms, which becomes critical in subsequent sections to infer TCRs and learn the hidden dynamics in the PIROM. The following discussion introduces three different approaches of decreasing fidelity to modeling the hypersonic TPS system: (i) a

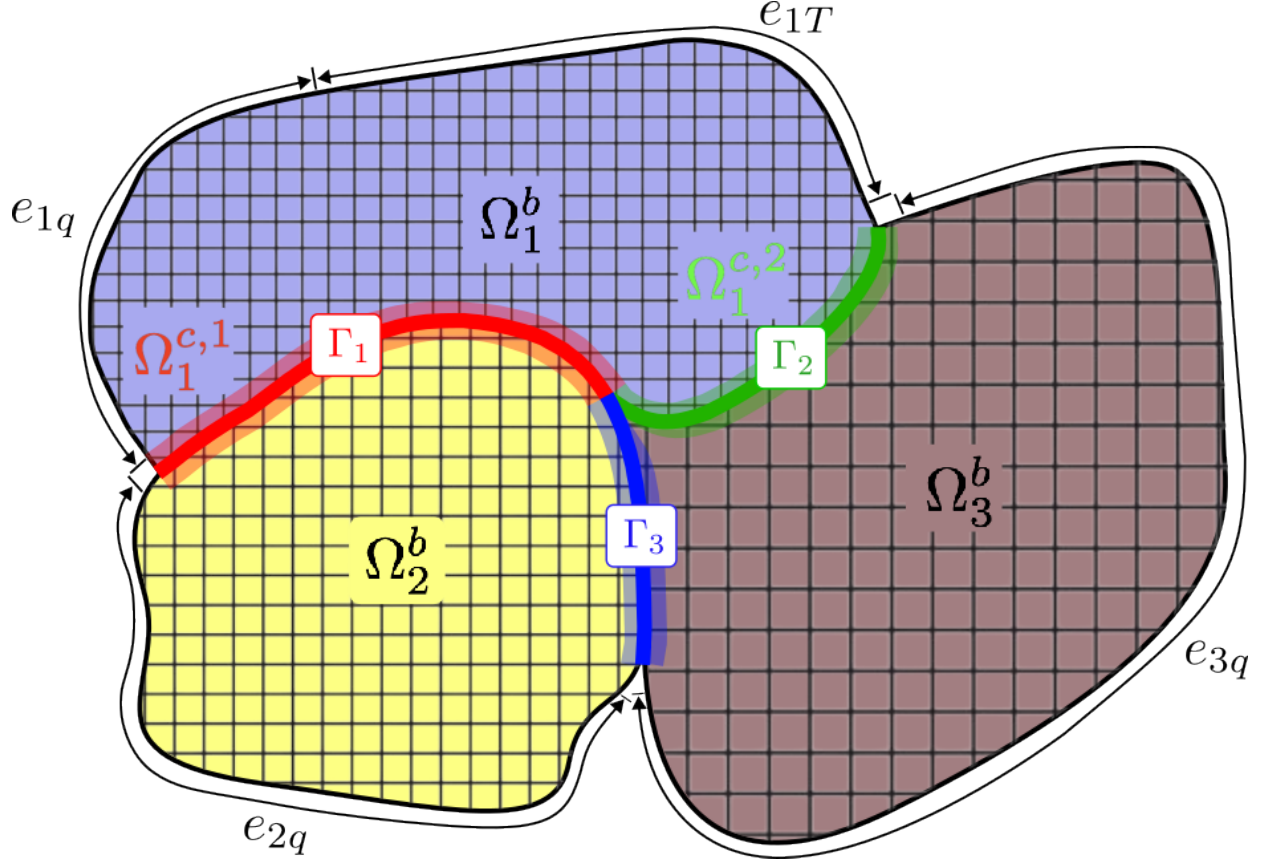


Figure 1: Schematic of the general TPS domain Ω composed of $N_C = 3$ components and $N_\Gamma = 3$ imperfect contact interfaces between components.

FOM based on a three-dimensional FEM, (ii) a coarse one-dimensional FEM, and (iii) a coarse finite-volume, i.e., lumped capacitance model (LCM).

2.2 Full-Order Finite-Element Model

The FOM is obtained by spatially discretizing the governing PDE using a Discontinuous Galerkin (DG) FEM, to result in a high-dimensional system of ODEs that model the transient temperature dynamics of the TPS. The DG-FEM formulation is leveraged for discretization as it: (i) provides a systematic framework for modeling multi-layered TPS with complex geometries, varying material properties, and interfacial TCRs, and (ii) is amenable to systematic model-order reduction via coarse-graining. Note that the high-fidelity numerical simulations in Sec. [X] are performed using standard FEM instead, and the equivalence between DG and standard FEM is noted upon their convergence.

Consider a DG-FEM discretization where the entire domain Ω is partitioned into non-overlapping bulk $\{B_i\}_{i=1}^B$ and interfacial $\{W_i\}_{i=1}^K$ elements, so that $\{E_i\}_{i=1}^M$ with $M = K + B$ corresponds to the complete collection of elements, and each element belongs to one and only one component in the TPS partition. Moreover, the perfect, imperfect, Neumann, and Dirichlet boundary interfaces are denoted as e_{ij} , Γ_{ij} , e_{iq} , and e_{iT} , respectively. Lastly, for an element E_i , $|\partial E_i|$ denotes the length ($d = 2$) or area ($d = 3$) of a boundary ∂E_i . Now, for the i -th element, use a set of P basis functions,

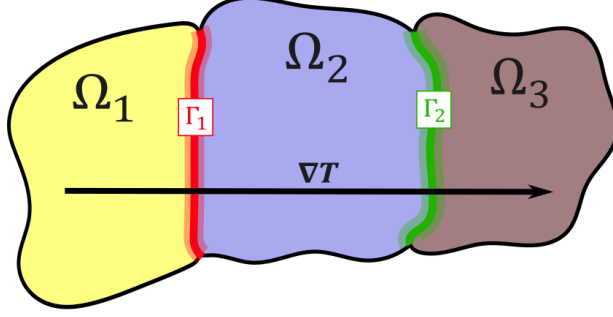


Figure 2: Schematic of a one-dimensional TPS domain Ω consisting of $N_C = 3$ components linearly discretized with only $N_I = 2$ imperfect interfaces. check comment

such as polynomials, to represent the temperature field as, 3

$$T_i(r, t) = \sum_{p=1}^P \phi_p(r) x_{i,p}(t), \quad r \in E_i \quad (2)$$

where $\{x_{i,p}\}_{p=1}^P$ corresponds to the states of the i -th element. Without loss of generality, let the basis functions on element E_i satisfy the following orthogonality condition,

$$\int_{E_i} \phi_p^i \phi_q^i d\Omega = |E_i| \delta_{pq}, \quad p \neq q \quad (3)$$

where δ_{pq} is the Kronecker delta. Furthermore, to aid in subsequent model-order reduction derivations, choose $\phi_1^i = 1$; therefore, by orthogonality, $\int_{E_i} \phi_p^i d\Omega = 0$ for $p > 1$. The first basis function therefore captures the volume-averaged temperature of the element, while the remaining basis functions capture the spatially varying temperature distributions within the element.

Using standard variational procedures, e.g., [Pernet2018], as detailed in Appendix [X], the full governing equation is denoted as,

$$\mathcal{A}(\mathbf{x}) \dot{\mathbf{x}} = \mathcal{B}(\mathbf{x}) \mathbf{x} + \mathcal{F}(t) \quad (4)$$

where $\mathbf{x} = [\mathbf{u}, \mathbf{w}]^\top \in \mathbb{R}^{MP}$ with $M = B + K$ includes the states for B bulk and K contact elements, i.e., $\mathcal{A} \in \mathbb{R}^{MP \times MP}$ and $\mathcal{B} \in \mathbb{R}^{MP \times MP}$ are the temperature-dependent mass and stiffness matrices, and $\mathcal{F}(t)$ corresponds to the contributions from the boundary conditions, respectively.

2.3 One-Dimensional Finite-Element Model

Next, a *one-dimensional* FEM (1D-FEM) ROM for the TPS is presented. This model assumes heat transfer is *one dimensional* along the TPS thickness, where the space-time varying temperature inside each layer is described using linear basis functions over a single finite element. Such 1D-FEM reduced-order modeling approach is commonly adopted in the literature due to the development of large temperature gradients along the thickness direction of the TPS, relative to in-plane temperature gradients [Gao2026, X]. Figure 2 depicts an example where $N_C = 3$ components are connected in series with $N_I = 2$ imperfect contacts.

Generalizing for a TPS containing M non-overlapping elements — still expressed with a coarse one-dimensional assumption — then yields

$$\mathbf{A}(\bar{\mathbf{u}}) \dot{\mathbf{u}} = \mathbf{B}(\bar{\mathbf{u}}) \mathbf{u} + \mathbf{F}(t) \quad (5)$$

where $\mathbf{A}, \mathbf{B} \in \mathbb{R}^{2M \times 2M}$ are block diagonals containing the mass and stiffness matrices for each element, and $\mathbf{F}$ is the forcing vector upholding the boundary condition at each edge. In a TPS system with perfect contacts, these components are solely products of the choice in basis function; for LBFs, they are defined as

$$\mathbf{A}_i(\bar{u}) = \rho c_p(\bar{u}) \frac{L}{6} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \quad (6a)$$

$$\mathbf{B}_i(\bar{u}) = \frac{k(\bar{u})}{L} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \quad (6b)$$

$$\mathbf{F} = \int_{\partial\Omega} \phi q_b d\Gamma \quad (6c)$$

where $\mathbf{A}_i, \mathbf{B}_i$ are the i -th blocks of $\mathbf{A}, \mathbf{B}$ respectively, and $\mathbf{F}$ is defined for the entire domain via its vector of LBFs, ϕ .

For a TPS with imperfect contacts, the bulk stiffness matrix in ?? no longer captures the total governing conductance of the TPS system, and thus must also account for the thermal resistances introduced by the imperfect contacts through the following heat interface heat flux condition,

$$\mathbf{B} = \underbrace{\mathbf{B}^b}_{\text{Satisfied}} + \underbrace{\mathbf{B}^c}_{\text{Missing}} \quad (7)$$

where the superscripts b and c differentiate the stiffness matrices that capture the *bulk* and *contact* heat fluxes, respectively. The choice in modeling the contact stiffness arises from the heat flux law

$$q = \frac{T_l - T_r}{R_x} \quad (8)$$

where the flux across a given element is defined by the change in temperature between the left and right sides over its thermal resistance. Leveraging the relationship of conductance and resistance, or

$$C = \frac{1}{R_x} \quad (9)$$

Eq. (8) can be rewritten as

$$q = C (T^- - T^+) \quad (10)$$

where T^- and T^+ represent the left and right sides of a given material block, respectively. Finally, generalizing for the full set of interactions between two locations on either side of a contact, the contact stiffness becomes

$$\mathbf{B}_j^c = C_j \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \quad (11)$$

for the j -th contact, where $j = i - 1$ due to the linearity in alignment of the elements (due to the one-dimensional assumption). The contact matrix can also be organized into a block diagonal matrix of shape $\mathbf{B}^c \in \mathbb{R}^{2M \times 2M}$, whose general structure is

$$\mathbf{B}^c = \begin{bmatrix} 0 & 0 & \cdots & 0 & 0 \\ 0 & \mathbf{B}_1^c & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & \mathbf{B}_j^c & 0 \\ 0 & 0 & \cdots & 0 & 0 \end{bmatrix} \quad (12)$$

as there is no contact interface on the edges of the TPS.

Expanding Eq. (5) to now encompass all aspects of the distant contact problem yields

$$\mathbf{A}(\bar{\mathbf{u}})\dot{\mathbf{u}} = \mathbf{B}^b(\bar{\mathbf{u}})\mathbf{u} + \mathbf{B}^c\mathbf{u} + \mathbf{F}(t) \quad (13)$$

where $\mathbf{B}^c$ is not a function of average temperature or time, as conductance of an imperfect contact is assumed non-zero and constant.

2.4 Reduced-Physics Model

Next, the *lumped capacitance model* (LCM) under imperfect TPS layer contacts is presented as the RPM for the TPS. In this formulation, each layer is modeled as a lumped mass with a time-varying average temperature, where TCRs introduce additional physics such as intra-layer temperature discontinuities. To handle both bulk and contact heat transfer, the following formulation introduces the decomposition of the TPS layers into *bulk* and *contact* sub-domains, serving as the basis for subsequent coarse-graining derivations, TCR identifiability, and PIROM formulations.

Bulk Temperature Dynamics Leveraging our previous work in Ref. [VargasVenegas2026], the *bulk* temperature dynamics of the LCM is derived at the component level from a point of view of energy balance,

$$\int_{\Omega_i^b} \rho c_{p,i}(\bar{u}) \dot{\bar{u}}_i d\Omega = \sum_{j \in \mathcal{N}_i^b} \int_{e_{ij}} q_{ij}^b d\Gamma + \int_{e_{iq}} q_b d\Gamma + \int_{e_{iT}} q_T d\Gamma \quad (14a)$$

$$\bar{A}_i \dot{\bar{u}}_i = \sum_{j \in \mathcal{N}_i^b} \int_{e_{ij}} q_{ij}^b d\Gamma + \int_{e_{iq}} q_b d\Gamma + \int_{e_{iT}} q_T d\Gamma, \quad i = 1, 2, \dots, N_B \quad (14b)$$

where $\bar{A}_i(\bar{u}_i) = \int_{\Omega_i^b} \rho c_{p,i}(\bar{u}) d\Omega$, and q_{ij}^b , q_{ij}^c , q_{ib} , and q_{iT} correspond to the heat flux contributions due to neighboring elements with perfect (e_{ij}) contact, imperfect (Γ_{ij}) contact, Neumann boundary conditions (e_{iq}), and Dirichlet boundary conditions (e_{iT}), respectively. Moreover, $\mathcal{N}_i^b$ and $\mathcal{N}_i^c$ are the sets of neighboring elements across perfect and imperfect interfaces, respectively.

Interface Temperature Dynmaics Two equivalent approaches are presented in this work to formulate the temperature jump dynamics: (i) coarse-graining the DG-FEM model and applying a *zero-energy storage assumption* for the contact sub-domains as shown in [APPENDIX], and (ii) leveraging energy conservation at the interface level. For brevity, the second approach is adopted here, whereas the first approach is demonstrated in the [APPENDIX]. By energy conservation, *the bulk and contact heat fluxes are equal* at the imperfect interfaces,

$$q_r^b(t) = \frac{\bar{u}^- - \bar{u}^+}{S_r(\bar{\mathbf{u}}) + R_r} = \frac{(\bar{u}^- - \bar{u}^+) - \Delta \bar{w}_r}{S_r(\bar{\mathbf{u}})} = g_r \Delta \bar{w}_r = q_r^c(t) \quad (15)$$

where the contact temperature jumps $\Delta \bar{w}_r$ for $r = 1, 2, \dots, N_\Gamma$ depend on the TCR values. The heat flux continuity algebraic constraints in Eq. (15) are differentiated with respect to time, i.e., DAE index reduction, such that for each interface,

$$(1 + g_r S_r(\bar{\mathbf{u}})) \Delta \dot{\bar{w}}_r = (\dot{\bar{u}}^- - \dot{\bar{u}}^+) - \frac{(\bar{u}^- - \bar{u}^+) - \Delta \bar{w}_r}{S_r(\bar{\mathbf{u}})} \dot{S}_r(\bar{\mathbf{u}}), \quad r = 1, 2, \dots, N_\Gamma \quad (16)$$

Collecting equations from all imperfect interfaces, the following system of ODEs governs the average contact temperature jump,

$$[\mathbf{I} + \mathbf{G} \odot \mathbf{S}(\bar{\mathbf{u}})] \Delta \dot{\bar{\mathbf{w}}} = \mathbf{D} \dot{\bar{\mathbf{u}}} - \mathbf{G} \left(\Delta \bar{\mathbf{w}} \odot \dot{\mathbf{S}}(\bar{\mathbf{u}}) \right) \quad (17)$$

where the state $\Delta \bar{\mathbf{w}} \in \mathbb{R}^{N_\Gamma}$, conductance $\mathbf{G} \in \mathbb{R}^{N_\Gamma \times N_\Gamma}$ and bulk resistance $\mathbf{S}(\bar{\mathbf{u}}) \in \mathbb{R}^{N_\Gamma \times N_\Gamma}$ are defined as,

$$\mathbf{G} = \text{diag}(g_1, g_2, \dots, g_{N_\Gamma}) \quad (18a)$$

$$\mathbf{S}(\bar{\mathbf{u}}) = \text{diag}(S_1(\bar{\mathbf{u}}), S_2(\bar{\mathbf{u}}), \dots, S_{N_\Gamma}(\bar{\mathbf{u}})) \quad (18b)$$

while $\mathbf{D} \in \mathbb{R}^{N_\Gamma \times N_C}$ is the difference matrix that computes the lumped temperature differences across the interfaces, and $\odot$ denotes the Hadamard product.

Reduced-Physics Model Formulation At component i , the following assumptions are employed: (i) the bulk $\bar{u}_i$ and contact $\bar{w}_i$ average temperatures are uniform and equal to the spatial average over the their respective Ω_i^b and Ω_i^c sub-domains in component Ω_i , (ii) for neighboring components i and j under perfect contact, the bulk heat flux is defined as $q_{ij}^b = g_{ij}(\bar{\mathbf{u}}) (\bar{u}_i - \bar{u}_j)$. The term $g_{ij}(\bar{\mathbf{u}}) = 1/R_{ij}(\bar{\mathbf{u}})$ is the effective thermal conductance, with $R_{ij} = R_i + R_j$ and $R_{ij}(\bar{\mathbf{u}}) = \ell/k(\bar{\mathbf{u}})A$, where ℓ is the distance between lumped masses, $k(\bar{\mathbf{u}})$ is the temperature-dependent thermal conductivity, and A is the cross-sectional area of the interface. Similarly, the heat flux due to the Dirichlet boundary condition is given by $q_{iT} = g_{iT}(\bar{\mathbf{u}}) (\bar{u}_i - \bar{u}_T)$. Now for component i , and noting that $\mathcal{N}_i^b = \mathcal{N}_i^u \cup \mathcal{N}_i^w$, Eq. (14b) may be written as,

$$\bar{A}_i(\bar{u}_i) \dot{\bar{u}}_i = \sum_{j \in \mathcal{N}_i^u} \int_{e_{ij}} g_{ij}(\bar{\mathbf{u}}) (\bar{u}_i - \bar{u}_j) de + \sum_{j \in \mathcal{N}_i^w} \int_{e_{ij}} g_{ij}(\bar{\mathbf{u}}) (\bar{u}_i - \bar{w}_j) de + \int_{e_{iq}} q_b de + \int_{e_{iT}} q_T de \quad (19a)$$

$$= \sum_{j \in \mathcal{N}_i^b \cup \{T\}} g_{ij}(\bar{\mathbf{u}}) |e_{ij}| \bar{u}_i - \sum_{j \in \mathcal{N}_i^u} g_{ij}(\bar{\mathbf{u}}) |e_{ij}| \bar{u}_j - \sum_{j \in \mathcal{N}_i^w} g_{ij}(\bar{\mathbf{u}}) |e_{ij}| \bar{w}_j + |e_{iq}| \bar{q}_b - g_{iT}(\bar{\mathbf{u}}) |e_{iT}| \bar{T}_b \quad (19b)$$

$$= \sum_{j \in \mathcal{N}_i^b \cup \{T\}} \bar{B}_{ij}^i \bar{u}_i + \sum_{j \in \mathcal{N}_i^u} \bar{B}_{ij}^j \bar{u}_j + \sum_{j \in \mathcal{N}_i^w} \bar{B}_{ij}^j \bar{w}_j + \bar{f}_i^{(b)}(t) \quad (19c)$$

or,

$$\left(\int_{\Omega_i^b} \rho c_{p,i}(\bar{u}) d\Omega \right) \dot{\bar{u}}_i = \left(\sum_{j \in \mathcal{N}_i^b} \int_{e_{ij}} g_{ij}(\bar{\mathbf{u}}) (\bar{u}_i - \bar{u}_j) d\Gamma \right) \quad (20)$$

$$+ \sum_{j \in \mathcal{N}_i^c} \left(\int_{\Gamma_{ij}} g_{x,ij} \Delta \bar{w}_j d\Gamma \right) + \int_{e_{iq}} q_b d\Gamma + \int_{e_{iT}} q_T d\Gamma \quad (21)$$

Collecting N_B bulk and N_Γ interface terms, the system of equations governing the coarse-grained temperature dynamics of the TPS with imperfect thermal contacts is written as the following coupled system of ODEs,

$$\bar{\mathcal{A}}(\mathbf{s}) \dot{\mathbf{s}} = \bar{\mathcal{B}}(\mathbf{s}) + \bar{\mathcal{F}}(t) \quad (22)$$

where,

$$\bar{\mathcal{A}} = \begin{bmatrix} \bar{\mathbf{A}}(\bar{\mathbf{u}}) & \mathbf{0} \\ -\mathbf{D} & \mathbf{I} + \mathbf{G} \odot \mathbf{S}(\bar{\mathbf{u}}) \end{bmatrix}, \quad \bar{\mathcal{B}} = \begin{bmatrix} \bar{\mathbf{B}}(\bar{\mathbf{u}}) & \bar{\mathbf{K}}(\bar{\mathbf{u}}) \\ \mathbf{0} & -\mathbf{G} \odot \dot{\mathbf{S}}(\bar{\mathbf{u}}) \end{bmatrix}, \quad \bar{\mathcal{F}}(t) = \begin{bmatrix} \bar{\mathbf{f}}(t) \\ \mathbf{0} \end{bmatrix} \quad (23)$$

and where $\mathbf{s} = [\bar{\mathbf{u}}, \Delta\bar{\mathbf{w}}]^\top \in \mathbb{R}^{N_C+N_\Gamma}$, and $\bar{\mathbf{A}}, \bar{\mathbf{B}} \in \mathbb{R}^{(N_C+N_\Gamma) \times (N_C+N_\Gamma)}$ define the RPM. Note that the $\bar{\mathbf{A}}$ and $\bar{\mathbf{B}}$ matrices include temperature dependence that represents the primary source of non-linearities for the problems considered in this work.

3 Physics-Infused Reduced-Order Modeling

The formulation of PIROM for TPS starts by mathematically connecting the FOM to the RPM. This connection is formally established through a coarse-graining procedure, which systematically reduces the spatial resolution of the DG-FEM to discover the lifted LCM model with contact temperature jump dynamics. Additionally, the coarse-graining procedure pinpoints residual terms in the RPM that facilitate the design of a data-driven memory closure to account for discarded spatial resolution.

3.1 Deriving the Reduced-Physics Model via Coarse-Graining

Following our previous work in Ref. [VargasVenegas2026], the LCM can be viewed as a zeroth-order (i.e., constant basis functions) DG-FEM with an extremely coarse mesh partition, and where the element-wise penalty factors become the conductance. In particular, the presence of imperfect interfaces in this work motivates modification of the coarse-graining procedure to explicitly account for TCR dynamics, which are not present in the standard LCM formulation. The following discussion presents the derivation of the lifted LCM dynamics via coarse-graining and the PIROM formulation to account for residual dynamics.

3.1.1 Coarse-Graining of States

Consider a DG-FEM formulation as in Eq. (4) with M elements, N_C components and N_Γ interfaces; clearly, $M \gg N_C, N_\Gamma$. Let component Ω_i contain a share a boundary with the imperfect interface Γ_r , i.e., $\partial\Omega_i^c \cap \Gamma_r \neq \emptyset$ with $\Omega_i^c \subset \Omega_i$, and define the sets of indices $\mathcal{V}_i$ and $\mathcal{F}_i$ for the elements in the bulk Ω_i^b and contact Ω_i^c sub-domains, respectively; note that $\Gamma_r = \partial\Omega_i^c \cap \partial\Omega_j^c$ with $i \neq j$, $i, j = 1, 2, \dots, N$ and $r = 1, 2, \dots, N_\Gamma$. Formally, the sets are defined as,

$$\mathcal{V}_i = \{j \mid E_j \subset \Omega_i^b\}, \quad i = 1, 2, \dots, N_B \quad (24a)$$

$$\mathcal{F}_i = \{j \mid \partial W_j \cap \Gamma_r \neq \emptyset, W_j \subset \Omega_i^c\}, \quad i = 1, 2, \dots, N, r = 1, 2, \dots, N_C \quad (24b)$$

Note that since an interface Γ_r is composed of two contact sub-domains, and there are N_C contact sub-domains, $N_C = 2N_\Gamma$. Moreover, to aid in subsequent RPM derivations, introduce the states for interfacing DG-FEM elements as,

$$\mathbf{w} = [\mathbf{w}_1^\top, \mathbf{w}_2^\top, \dots, \mathbf{w}_K^\top] = [\mathbf{w}^-, \mathbf{w}^+]^\top \in \mathbb{R}^{KP}$$

associated to the elements with neighboring imperfect interfaces on the negative and positive sides, respectively; these variables are coupled to the bulk element states $\mathbf{u} \in \mathbb{R}^{BP}$ via the standard DG-FEM penalty terms, as shown in Appendix [X]. Additionally, note that $M = B + K$, and $\mathbf{w}^- \in \mathbb{R}^{K^-P}$ and $\mathbf{w}^+ \in \mathbb{R}^{K^+P}$, with $K = K^- + K^+$ and where K^- and K^+ denote the total number of elements on the negative and positive sides of imperfect interfaces, respectively.

Under such variable construction, the average temperatures on some bulk component Ω_i^b and interface Γ_r are given by,

$$\bar{u}_i = \frac{1}{|\Omega_i^b|} \sum_{m \in \mathcal{V}_i} \int_{E_m} \phi_m^\top \mathbf{u}_m dE = \frac{1}{|\Omega_i^b|} \sum_{m \in \mathcal{V}_i} |E_m| \boldsymbol{\varphi}_m^{j^\top} \mathbf{u}_m, \quad i = 1, \dots, N_C \quad (25a)$$

$$\bar{w}_r^-(t) = \frac{1}{|\Omega_r^{c,-}|} \sum_{m \in \mathcal{F}_r^-} \int_{E_m} \phi_m^\top \mathbf{w}_m^- dE = \frac{1}{|\Omega_r^{c,-}|} \sum_{m \in \mathcal{F}_r^-} |E_m| \psi_m^{r-} \mathbf{w}_m^-, \quad r = 1, 2, \dots, N_\Gamma \quad (25b)$$

$$\bar{w}_r^+(t) = \frac{1}{|\Omega_r^{c,+}|} \sum_{m \in \mathcal{F}_r^+} \int_{E_m} \phi_m^\top \mathbf{w}_m^+ dE = \frac{1}{|\Omega_r^{c,+}|} \sum_{m \in \mathcal{F}_r^+} |E_m| \psi_m^{r+} \mathbf{w}_m^+, \quad r = 1, 2, \dots, N_\Gamma \quad (25c)$$

where the orthogonal basis functions $\varphi_i^j, \psi_m^{j\pm} = [1, 0, \dots, 0] \in \mathbb{R}^P$ if $i \in \mathcal{V}_j$ and $m \in \mathcal{F}_j^\pm$, respectively, and are zero otherwise.

Given the average temperatures $\{\bar{u}_i\}_{i=1}^{N_C}$, $\{\bar{w}_i^-\}_{i=1}^{N_\Gamma}$, and $\{\bar{w}_i^+\}_{i=1}^{N_\Gamma}$, the DG-FEM states of elements on domains Ω_i^b , $\Omega_r^{c,-}$, and $\Omega_r^{c,+}$ are written as,

$$\mathbf{u}_i = \sum_{k=1}^{N_C} \varphi_i^k \bar{u}_k + \delta \mathbf{u}_i, \quad i \in \mathcal{V}_j, \quad j = 1, 2, \dots, B \quad (26a)$$

$$\mathbf{w}_{ir}^- = \sum_{k=1}^{N_\Gamma} \psi_k^{r-} \bar{w}_k^- + \delta \mathbf{w}_{ir}^-, \quad i \in \mathcal{F}_r^-, \quad r = 1, 2, \dots, N_\Gamma \quad (26b)$$

$$\mathbf{w}_{ir}^+ = \sum_{k=1}^{N_\Gamma} \psi_k^{r+} \bar{w}_k^+ + \delta \mathbf{w}_{ir}^+, \quad i \in \mathcal{F}_r^+, \quad r = 1, 2, \dots, N_\Gamma \quad (26c)$$

where $\varphi_i^k, \psi_i^{k\pm} = [1, 0, \dots, 0]^\top \in \mathbb{R}^P$ if $i \in \mathcal{V}_k$ and $\mathcal{F}_r^\pm$, respectively.

The Eqs. (26a) to (26c) are collected in matrix form,

$$\mathbf{u} = \Phi \bar{\mathbf{u}} + \delta \mathbf{u}, \quad \bar{\mathbf{u}} = \Phi^\dagger \mathbf{u} \quad (27a)$$

$$\mathbf{w} = \Pi \bar{\mathbf{w}} + \delta \mathbf{w}, \quad \bar{\mathbf{w}} = \Pi^\dagger \mathbf{w} \quad (27b)$$

where $\Phi \in \mathbb{R}^{BP \times NB}$ is a matrix of $B \times N_B$ blocks, with the (i, j) -th block as φ_i^j , $\Phi^\dagger \in \mathbb{R}^{N_C \times BP}$ is the left inverse of Φ , with the (i, j) -th block as $\varphi_i^{j\dagger} = \frac{|E_i|}{|\Omega_j^c|} \varphi_i^{jT}$ and $\delta \mathbf{u}$ is the collection of deviations from the average temperature. Moreover,

$$\Pi = \begin{bmatrix} \mathbf{M} & \mathbf{0} \\ \mathbf{0} & \mathbf{N} \end{bmatrix} \in \mathbb{R}^{KP \times NC}, \quad \Pi^\dagger = \begin{bmatrix} \mathbf{M}^\dagger & \mathbf{0} \\ \mathbf{0} & \mathbf{N}^\dagger \end{bmatrix} \in \mathbb{R}^{2N_\Gamma \times KP} \quad (28)$$

where $\mathbf{M} \in \mathbb{R}^{K \times P \times N_\Gamma}$ and $\mathbf{N} \in \mathbb{R}^{K \times P \times N_\Gamma}$ correspond to matrices that project the average interface temperatures $\bar{\mathbf{w}} = [\bar{\mathbf{w}}^-, \bar{\mathbf{w}}^+] \in \mathbb{R}^{2N_\Gamma}$ to the full-order interface DG-FEM state. Therefore, $\Pi \in \mathbb{R}^{KP \times 2N_\Gamma}$ is a matrix of $K \times 2N_\Gamma$ blocks with the (i, j) -th block as ψ_i^j , $\Pi^\dagger \in \mathbb{R}^{2N_\Gamma \times KP}$ is the left inverse of Π with the (i, j) -th block as $\psi_i^{j\dagger} = \frac{|E_i|}{|\Omega_j^c|} \psi_i^{jT}$ and $\delta \mathbf{w}$ is the collection of deviations from the interfacial average temperatures. By definition, the matrices defined above satisfy,

$$\begin{aligned} \Phi^\dagger \Phi &= \mathbf{I}, & \Phi^\dagger \delta \mathbf{u} &= \mathbf{0} \\ \Pi^\dagger \Pi &= \mathbf{I}, & \Pi^\dagger \delta \mathbf{w} &= \mathbf{0} \end{aligned}$$

For brevity, the states in Eq. (27b) are collectively written as,

$$\mathbf{x} = \mathbf{X} \bar{\mathbf{x}} + \delta \mathbf{x}, \quad \bar{\mathbf{x}} = \mathbf{X}^\dagger \mathbf{x} \quad (30)$$

where $\mathbf{x} = [\mathbf{u}, \mathbf{w}]^\top \in \mathbb{R}^{MP}$, $\mathbf{X} \in \mathbb{R}^{MP \times (N_B + N_C)}$, and $\mathbf{X}^\dagger \in \mathbb{R}^{(N_B + N_C) \times MP}$.

3.1.2 Coarse-Graining of Dynamics

Next, the coarse-graining procedure follows our previous work in Ref. [VargasVenegas2026], where the higher-order temperature modes are neglected in the DG-FEM dynamics, leading to a spatially averaged transient modeling of the TPS temperature dynamics. The presence of imperfect thermal contacts introduces additional modeling considerations to account for temperature discontinuities across TPS layers. The following discussion summarizes the critical results from coarse-graining, while detailed derivations are provided in Appendix [X].

Consider a function of the states in the form $\mathbf{M}(\mathbf{x})\mathbf{g}(\mathbf{x})$ with $\mathbf{x} = [\mathbf{u}, \mathbf{w}]^\top \in \mathbb{R}^{MP}$ and $M = B + K$, where $\mathbf{g}(\mathbf{x}) : \mathbb{R}^{MP} \rightarrow \mathbb{R}^{MP}$ is a vector-valued function, and $\mathbf{M}(\mathbf{x}) : \mathbb{R}^{MP} \rightarrow \mathbb{R}^{p \times MP}$ is a matrix-valued function with an arbitrary dimension p . Define the projection matrices $\mathbf{P} = \mathbf{X}\mathbf{X}^\dagger$ where,

$$\mathbf{X} = \text{diag}(\Phi, \Pi) \in \mathbb{R}^{MP \times (N_B + N_C)} \quad (31a)$$

$$\mathbf{X}^\dagger = \text{diag}(\Phi^\dagger, \Pi^\dagger) \in \mathbb{R}^{(N_B + N_C) \times MP} \quad (31b)$$

and the projection operator,

$$\mathcal{P}[\mathbf{M}(\mathbf{x})\mathbf{g}(\mathbf{x})] = \mathbf{M}(\mathbf{P}\mathbf{x})\mathbf{P}\mathbf{g}(\mathbf{P}\mathbf{x}) \quad (32a)$$

$$= \mathbf{M}(\mathbf{X}\mathbf{X}^\dagger\mathbf{x})\mathbf{X}\mathbf{X}^\dagger\mathbf{g}(\mathbf{X}\mathbf{X}^\dagger\mathbf{x}) \quad (32b)$$

$$= \mathbf{M}(\mathbf{X}\bar{\mathbf{x}})\mathbf{X}\mathbf{X}^\dagger\mathbf{g}(\mathbf{X}\bar{\mathbf{x}}) \quad (32c)$$

so that the resulting function depends only on the resolved variables $\bar{\mathbf{x}} = [\bar{\mathbf{u}}, \bar{\mathbf{w}}^-, \bar{\mathbf{w}}^+]^{N_B + N_C}$; note that an averaged temperature difference $\Delta\bar{\mathbf{w}} = \bar{\mathbf{w}}^- - \bar{\mathbf{w}}^+$ may be computed between two contact sub-domains conforming an imperfect interface, which becomes useful for subsequent TCR identifications. Correspondingly, the residual operator is $\mathcal{Q} = \mathcal{I} - \mathcal{P}$ so that $\mathcal{Q}[\mathbf{M}(\mathbf{x})\mathbf{g}(\mathbf{x})] = \mathbf{M}(\mathbf{x})\mathbf{g}(\mathbf{x}) - \mathcal{P}[\mathbf{M}(\mathbf{x})\mathbf{g}(\mathbf{x})]$. When the function is not separable, the projection operator is simply given by $\mathcal{P}[\mathbf{g}(\mathbf{x})] = \mathbf{g}(\mathbf{X}\bar{\mathbf{x}})$.

Subsequently, the operators defined above are applied to coarse-grain the dynamics. First, the DG-FEM model in Eq. (4) is written as,

$$\dot{\mathbf{x}} = \mathcal{A}^{-1}[\mathcal{B}(\mathbf{x})\mathbf{x} + \mathcal{F}(t)] = \mathbf{r}(\mathbf{x}, t) \quad (33)$$

Multiplying both sides by $\mathbf{X}^\dagger$ produces,

$$\mathbf{X}^\dagger\dot{\mathbf{x}} = \mathbf{X}^\dagger(\mathbf{X}\dot{\bar{\mathbf{x}}} + \delta\dot{\mathbf{x}}) = \dot{\bar{\mathbf{x}}} = \mathbf{X}^\dagger\mathbf{r}(\mathbf{x}, t) \quad (34)$$

and applying the projection and residual operator yields,

$$\dot{\bar{\mathbf{x}}} = \mathcal{P}[\mathbf{X}^\dagger\mathbf{r}(\mathbf{x}, t)] + \mathcal{Q}[\mathbf{X}^\dagger\mathbf{r}(\mathbf{x}, t)] \quad (35a)$$

$$= \mathbf{r}^{(1)}(\bar{\mathbf{x}}, t) + \mathbf{r}^{(2)}(\mathbf{x}, t) \quad (35b)$$

where $\mathbf{r}^{(1)}(\bar{\mathbf{x}}, t)$ is the resolved dynamics, which depends only on $\bar{\mathbf{x}}$, and $\mathbf{r}^{(2)}(\mathbf{x}, t)$ is the residual dynamics, which depends on the full-order state $\mathbf{x}$. Detailed derivations of the resolved $\mathbf{r}^{(1)}$ and residual $\mathbf{r}^{(2)}$ dynamics are provided in Appendix [X], and the key results are stated below.

First, the resolved dynamics becomes exactly the LCM, with the temperature jump dynamics formulated as a *zero-energy storage* assumption on the contact sub-domain component of the DG-FEM dynamics. Specifically, the coarse-grained mass matrices are given by,

$$\bar{\mathbf{A}} = \mathbf{W}_u \left(\Phi^\dagger \mathbf{A} (\Phi \bar{\mathbf{u}})^{-1} \Phi \right)^{-1} \quad (36a)$$

$$\bar{\mathbf{D}} = \mathbf{W}_w \left(\mathbf{\Pi}^\dagger \mathbf{D} (\mathbf{\Pi} \bar{\mathbf{w}})^{-1} \mathbf{\Pi} \right)^{-1} \quad (36b)$$

where $\mathbf{W}_u \in \mathbb{R}^{N_B \times N_B}$ and $\mathbf{W}_w \in \mathbb{R}^{N_C \times N_C}$ are diagonal weight matrices such that $[\mathbf{W}_u]_i = |\mathcal{V}_k|$ if $i \in \mathcal{V}_k$, and $[\mathbf{W}_c]_i = |\mathcal{F}_k|$ if $i \in \mathcal{F}_k$. Similarly, the coarse-grained stiffness matrices are,

$$\bar{\mathbf{B}} = \mathbf{W}_u \mathbf{\Phi}^\dagger \mathbf{B} (\mathbf{\Phi} \bar{\mathbf{u}}) \mathbf{\Phi} \quad (37a)$$

$$\bar{\mathbf{C}} = \mathbf{W}_u \mathbf{\Phi}^\dagger \mathbf{C} (\mathbf{\Phi} \bar{\mathbf{u}}) \mathbf{\Pi} \quad (37b)$$

$$\bar{\mathbf{E}} = \mathbf{W}_w \mathbf{\Pi}^\dagger \mathbf{E} (\mathbf{\Phi} \bar{\mathbf{u}}) \mathbf{\Phi} \quad (37c)$$

$$\bar{\mathbf{F}} = \mathbf{W}_w \mathbf{\Pi}^\dagger \mathbf{F} (\mathbf{\Phi} \bar{\mathbf{u}}) \mathbf{\Pi} \quad (37d)$$

where

3.2 Formulation of Reduced-Order Model

The Mori-Zwanzig (MZ) formalism is an operator-projection technique used to derive ROMs from high-dimensional dynamical systems, especially in statistical mechanics and fluid dynamics [Parish2017, Stinis2017, Parish2025]. The MZ formalism enables an exact reformulation of the FOM in terms of a subset of resolved states, with unresolved dynamical terms appearing as residual

The proposed PIROM is subsequently developed based on such reformulation.

4 Bulk-only Mori–Zwanzig Memory Closure

The purpose of the closure is to account for the unresolved dynamics in the LCM, while leaving the contact-jump dynamics unchanged. Accordingly, the

Accordingly, write the Projected Generalized Langevin Equation (PGLE) for the lumped-temperature dynamics,

$$A(\mathbf{u}) \dot{\mathbf{u}} = \sum_{i=1}^{N-1} \frac{1}{S_i(\mathbf{u})} (\mathbf{K}_i u + \mathbf{L}_i \Delta \boldsymbol{\theta}) + \mathbf{f}(t) + \int_{t_0}^t \boldsymbol{\kappa}(t, s, u, \mathbf{f}; \Theta) ds \quad (38)$$

Now, ensure the designed memory modifies only the bulk conductances, i.e.,

$$\boldsymbol{\kappa} \in \text{span} \{ \mathbf{K}_1 u(t), \mathbf{K}_2 u(t), \mathbf{K}_3 u(t), \mathbf{L}_1 \Delta \boldsymbol{\theta}_1(t), \mathbf{L}_2 \Delta \boldsymbol{\theta}_2(t), \mathbf{L}_3 \Delta \boldsymbol{\theta}_3(t) \} \quad (39)$$

which leads to,

$$\boldsymbol{\kappa}(t, s, u, \mathbf{f}; \Theta) \approx \sum_{j=1}^m \kappa_j(t, s, u, \mathbf{f}; \Theta) \mathcal{P}_j \quad (40)$$

where,

$$\mathcal{P}_j = \sum_{i=1}^{N-1} \frac{1}{S_i(u)} \mathbf{p}_{ij} \quad (41)$$

Now, define the memory variables as,

$$\beta_j(t) = \int_{t_0}^t \kappa_j(t, s, u, \mathbf{f}; \Theta) ds \quad (42)$$

then,

$$\int_{t_0}^t \boldsymbol{\kappa}(t, s, u, \mathbf{f}; \Theta) ds = \sum_{j=1}^m \beta_j(t) \mathcal{P}_j \quad (43)$$

It follows that,

$$\sum_{j=1}^m \beta_j(t) \mathbf{P}_j = \sum_{j=1}^m \beta_j(t) \sum_{i=1}^{N-1} \frac{1}{S_i(u)} \mathbf{P}_{ij} = \sum_{i=1}^{N-1} \frac{1}{S_i(u)} \left(\sum_{j=1}^m \mathbf{P}_{ij} \beta_j(t) \right) = \sum_{i=1}^{N-1} \frac{1}{S_i(u)} \mathbf{P}_i \boldsymbol{\beta} \quad (44)$$

Then, the memory correction to the LCM dynamics leads to the following modified lumped-temperature dynamics,

$$A(\mathbf{u}) \dot{\mathbf{u}} = \sum_{i=1}^{N-1} \frac{1}{S_i(\mathbf{u})} (\mathbf{K}_i u + \mathbf{L}_i \Delta \boldsymbol{\theta} + \mathbf{P}_i \boldsymbol{\beta}) + \mathbf{f}(t) \quad (45)$$

Now, let,

$$\kappa_i(t, s, \mathbf{u}, \mathbf{f}; \Theta) = e^{-\lambda_j(t-s)} \left(q_j^\top \phi(\mathbf{u}(s)) + r_j^\top \psi(\mathbf{f}(s)) \right) \quad (46)$$

such that the memory variables are given by,

$$\boldsymbol{\beta}(t) = \int_{t_0}^t e^{-\Lambda(t-s)} \left(Q^\top \phi(\mathbf{u}(s)) + R^\top \psi(\mathbf{f}(s)) \right) ds \quad (47)$$

The hidden dynamics are now given by,

$$\dot{\boldsymbol{\beta}} = -\Lambda \boldsymbol{\beta} + Q^\top \phi(\mathbf{u}) + R^\top \psi(\mathbf{f}) \quad (48)$$

and in state-space form, the hidden dynamics are,

$$\dot{\boldsymbol{\beta}} = -\Lambda \boldsymbol{\beta} + Q^\top \phi(\mathbf{u}) + R^\top \psi(\mathbf{f}) \quad (49)$$

The full lifted and closed system is given by,

$$A(\mathbf{u}) \dot{\mathbf{u}} = \sum_{i=1}^{N-1} \frac{1}{S_i(\mathbf{u})} (\mathbf{K}_i \mathbf{u} + \mathbf{L}_i \Delta \boldsymbol{\theta}) + \mathbf{f}(t) \quad (50a)$$

$$M(\mathbf{u}; R_x) \dot{\Delta \boldsymbol{\theta}} = \mathbf{D} \dot{\mathbf{u}} - G_x(\Delta \boldsymbol{\theta} \odot \dot{S}(\mathbf{u})) \quad (50b)$$

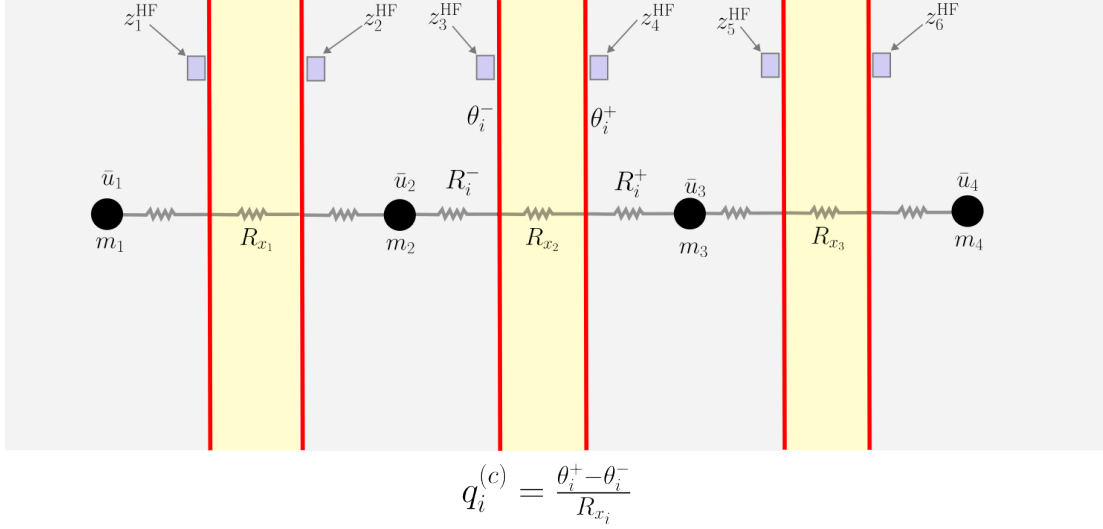
$$\dot{\boldsymbol{\beta}} = -\Lambda \boldsymbol{\beta} + Q^\top \phi(\mathbf{u}) + R^\top \psi(\mathbf{f}) \quad (50c)$$

$$\mathbf{z} = \mathbf{H}_u \mathbf{u} + \mathbf{H}_\theta \Delta \boldsymbol{\theta} + \mathbf{H}_\beta \boldsymbol{\beta} \quad (50d)$$

$$\mathcal{J} = \alpha_1 \mathcal{J}_{\text{TCR}} + \alpha_2 \mathcal{J}_\beta \quad (51)$$

5 Application to Thermal Protection Systems

In this section, the proposed PIROM approach is applied to modeling a four-layered hypersonic TPS with unknown contact resistances, and is compared against a representative physics-based model, i.e., FEMs, and data-driven surrogate model, i.e., a neural ordinary differential equation (NODE) model. The performance of the models is assessed in terms of their ability to infer the TCRs, and to predict the transient temperature dynamics under varying boundary conditions and material properties. The configuration of the TPS, problem parametrization, optimal model design for inference, and numerical results are presented next.



5.1 Problem Description and Parametrization

5.2 Model Selection

5.2.1 One-Dimensional Finite-Element Model

1. Discuss sensor placement and how it is done for this specific system (i.e. LBFs)
2. Tradeoff for C vs δ vs SNR (however, this might be best placed in the appendix with the main takeaways in the main document)

5.2.2 Lumped Capacitance Model

1. Discuss sensor placement and how it is done for this specific system (i.e. moving indices in FEM)
2. Tradeoff for C vs δ vs SNR (however, this might be best placed in the appendix with the main takeaways in the main document)

5.2.3 Remarks

1. What model won? E.g. Model A with Noise B performed the best
2. All else should perhaps be placed in the appendix, with this section only highlighting what model we will compare the PIROM to (to reduce total number of combinations and overall noise in the paper)

5.2.4 Model Performance on Transient-Dominated Timescale

5.3 PIROM Results

5.3.1 PIROM Optimization

1. Hidden dynamics
2. TCR

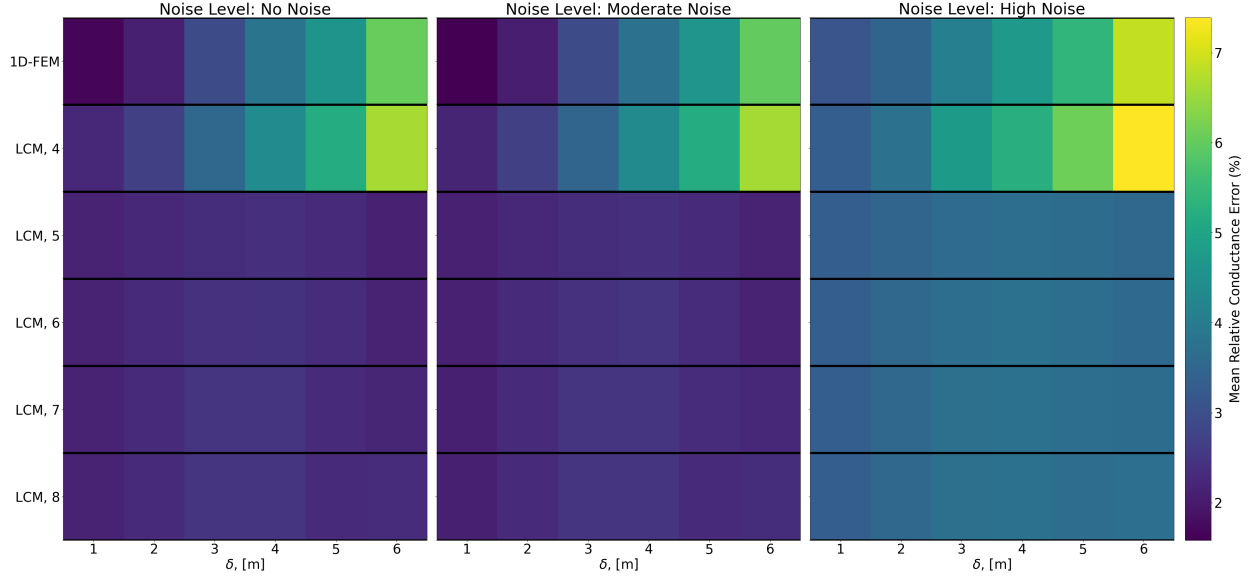


Figure 3: .

3. Training costs

5.3.2 BC Generalization

5.3.3 Material Generalization

5.3.4 TCR Generalization

6 Concluding Remarks

The main conclusion of this work are as follows,

1. A one-way physics-data PIROM design is found to successfully infer the unknown TCRs and hidden dynamics simultaneously. During inference, X and Y orders of magnitude in *training efficiency* are obtained relative to the FOM and 1D-FEM forward models, while X and Y orders of magnitude in *evaluation efficiency*

X and Y orders of magnitude in *training efficiency* are obtained relative to the FOM and 1D-FEM based model, while X and Y orders of magnitude in *evaluation efficiency*

successfully infers the unknown TCRs and hidden dynamics simultaneously, offering X and Y orders of magnitude in *training efficiency*, as well as X and Y orders of magnitude in *evaluation efficiency*

relative to the FOM and 1D-FEM based model, while X and Y orders of magnitude in *evaluation efficiency*

and X and Y orders of magnitude in *evaluation efficiency*

TCRs from noisy thermocouple

framework successfully learns a ROM for the prediction of the temperature dynamics of the TPS, while simultaneously inferring the unknown contact conductances with errors of less than X% on noisy data with SNR of up to Y dB.

2. The PIROM outperforms the state-of-the-art data-driven, NN-based SciML, and traditional FEM modeling approaches for inferring contact conductances, as well as for predicting the temperature dynamics for OOD configurations.
3. Model bias is found to significantly affect the

A Appendix

This section provides the technical details for the main text, including the DG-FEM formulation of the FOM, as well as the coarse-graining procedure to derive the LCM with interface temperature jump dynamics.

A.1 Discontinuous Galerkin Finite-Element Formulation

Domain Discretization Consider the partition of the TPS spatial domain into N_C continuous components $\{\Omega_i\}_{i=1}^{N_C}$, where each component has its own temperature-dependent material properties. The domain of each component Ω_i is further divided into two sub-domains: (i) $\Omega_i^c \subset \Omega_i$ consisting of a thin layer of material that *is in direct contact with neighboring components across an imperfect interface*, and (ii) $\Omega_i^b \subset \Omega_i$ consisting of the *remaining bulk material that does not have boundaries in direct contact with neighboring components*. Note that one component may have multiple imperfect-contact interfaces, and thus may have multiple contact sub-domains Ω_i^c . For example, in Fig. 1 the domain Ω_2 has two neighboring components with imperfect contacts over interfaces Γ_1 and Γ_3 , and thus has two contact sub-domains $\Omega_2^{c,1}$ and $\Omega_2^{c,3}$, and one bulk sub-domain Ω_2^b ; clearly, $\Omega_2 = \Omega_2^{c,1} \cup \Omega_2^{c,3} \cup \Omega_2^b$ and $\Omega_2^{c,1} \cap \Omega_2^{c,3} = \emptyset$.

Variational Formulation Choosing appropriate basis functions $v(x)$ and $w(x)$, the variational formulation of the governing equations is obtained element-wise for each component in the TPS. To aid in the subsequent model derivations, denote $\mathcal{N}_i$ as the collection of neighboring elements to E_i , which are split into,

$$\mathcal{N}_i = \mathcal{N}_i^b \cup \mathcal{N}_i^c, \quad i = 1, \dots, M \quad (52)$$

where $\mathcal{N}_i^b = \mathcal{N}_i^u \cup \mathcal{N}_i^w$ includes the indices of neighboring elements on the Ω_i^b and Ω_i^c sub-domains *across perfect interfaces*. An additional set of neighboring elements *across imperfect interfaces* $\mathcal{N}_i^c$ is introduced. Both sets are formally defined as,

$$\mathcal{N}_i^b = \{j \in \{1, \dots, M\} \mid E_j \cap E_i = e_{ij} \neq \emptyset, e_{ij} \subseteq \Omega\}, \quad (53a)$$

$$\mathcal{N}_i^c = \{j \in \{1, \dots, M\} \mid \partial E_i \cap \partial E_j \cap \Gamma_r \neq \emptyset\}, \quad r = 1, 2, \dots, N_\Gamma \quad (53b)$$

where $\{\Gamma_r\}_{r=1}^{N_\Gamma}$ correspond to the collection of imperfect interfaces. Under this separation, the Interior Penalty Galerkin (IPG) scheme yields the following bi-linear form for the variational formulation of the governing equations [Cohen2018],

$$\sum_{i=1}^M [a_{1,i}(v, w) + a_{2,i}(v, w) + a_{3,i}(v, w)] = \sum_{i=1}^M L_i(v) \quad (54)$$

where ϵ is a user-specified parameter, and,

$$a_{1,i}(v, w) = \int_{E_i} \left(\rho_i c_{p,i} v \frac{\partial w}{\partial t} + \mathbf{k}_i \nabla v \cdot \nabla w \right) dE_i \quad (55a)$$

$$\begin{aligned}
a_{2,i}(v, w) = & - \sum_{j \in \mathcal{N}_i^b \cup \{T\}} \int_{e_{ij}} \{\mathbf{k}_i \nabla v \cdot \mathbf{n}_i\} [w] de_{ij} + \epsilon \sum_{j \in \mathcal{N}_i^b \cup \{T\}} \int_{e_{ij}} \{\mathbf{k}_i \nabla w \cdot \mathbf{n}_i\} [v] de_{ij} \\
& + \sigma_b \sum_{j \in \mathcal{N}_i^b \cup \{T\}} \int_{e_{ij}} [v] [w] de_{ij}
\end{aligned} \tag{55b}$$

$$\begin{aligned}
a_{3,i}(v, w) = & - \sum_{j \in \mathcal{N}_i^c} \int_{\Gamma_{ij}} \{\mathbf{k}_i \nabla v \cdot \mathbf{n}_i\} [w] d\Gamma + \epsilon \sum_{j \in \mathcal{N}_i^c} \int_{\Gamma_{ij}} \{\mathbf{k}_i \nabla w \cdot \mathbf{n}_i\} [v] d\Gamma \\
& + \sigma_c \sum_{j \in \mathcal{N}_i^c} \int_{\Gamma_{ij}} [v] [w] d\Gamma
\end{aligned} \tag{55c}$$

$$L_i(v) = \epsilon \sum_{j \in \mathcal{N}_i^b \cup \{T\}} \int_{e_{ij}} (\mathbf{k}_i \nabla w \cdot \mathbf{n}_i) T_b de_{ij} + \int_{e_{iq}} v q_b de_{iq} + \sigma \int_{e_{iT}} v T_b de_{iT} \tag{55d}$$

where $a_{1,i}$, $a_{2,i}$, $a_{3,i}$, and $L_i(v)$ amount to heat conduction due to bulk energy transfer, between-element perfect interfaces e_{ij} , between-element imperfect interfaces Γ_{ij} , and boundary conditions, respectively. In the bi-linear forms, the notations of $[]$ and $\{\}$ are respectively the jumps and averages at an interface e_{ij} shared by two neighboring elements E_i and E_j , defined as,

$$[u] = u|_{E_i} - u|_{E_j}, \quad \{u\} = \frac{1}{2} \left(u|_{E_i} + u|_{E_j} \right), \quad x \in e_{ij} = \partial E_i \cap \partial E_j \tag{56}$$

where the same holds for an imperfect interface Γ_{ij} , and where σ_b, σ_c are known user-specified bulk and interface penalty terms whose value depends on the element size. Depending on the choice of ϵ , the bi-linear form may correspond to symmetric IPG ($\epsilon = -1$), non-symmetric IPG ($\epsilon = 1$), or incomplete IPG ($\epsilon = 0$) schemes. All these schemes are consistent with the original PDE and have a similar convergence rate with respect to mesh size.

Galerkin Projections Next, the DG-FEM is written in an element-wise form. For the i -th element, use a set of P basis functions, such as polynomials, to represent the temperature field as in Eq. (2). Using the same test functions as trial functions, the DG-FEM dynamics for the i -th element is obtained by evaluating the element-wise bi-linear forms,

$$a_{1,i}(\phi_k^i, T) + a_{2,i}(\phi_k^i, T) + a_{3,i}(\phi_k^i, T) = L_i(\phi_k^i), \quad k = 1, \dots, P \quad i = 1, 2, \dots, M \tag{57}$$

where ϕ_k^i is the k -th basis function for the i -th element. Evaluating for all basis functions, while leveraging the splitting of DG-FEM states into bulk and contact elements, the following system of coupled ODEs is obtained,

- **Bulk elements:** for $i = 1, \dots, B$,

$$\begin{aligned}
\mathbf{A}_i(\mathbf{u}_i) \dot{\mathbf{u}}_i = & \underbrace{\left[\mathbf{B}_i(\mathbf{u}_i) \mathbf{u}_i + \sum_{j \in \mathcal{N}_i^b} \mathbf{B}_{ij}^i \mathbf{u}_i + \sum_{j \in \mathcal{N}_i^u} \mathbf{B}_{ij}^j \mathbf{u}_j \right]}_{\text{bulk dynamics on } \Omega_i^b} + \underbrace{\left[\sum_{j \in \mathcal{N}_i^w} \mathbf{B}_{ij}^j \mathbf{w}_j \right]}_{\text{coupling to dynamics on } \Omega_i^c} + \underbrace{\mathbf{f}_i^{(b)}(t)}_{\text{heating BC}}
\end{aligned} \tag{58}$$

- **Imperfect interface elements:** for $i = 1, \dots, K$,

$$\begin{aligned}
\mathbf{D}_i(\mathbf{w}_i) \dot{\mathbf{w}}_i = & \underbrace{\left[\sum_{j \in \mathcal{N}_i^u} \mathbf{B}_{ij}^j \mathbf{u}_j \right]}_{\text{coupling to dynamics on } \Omega_i^u} + \underbrace{\left[\mathbf{B}_i(\mathbf{w}_i) \mathbf{w}_i + \sum_{j \in \mathcal{N}_i^b} \mathbf{B}_{ij}^i \mathbf{w}_i + \sum_{j \in \mathcal{N}_i^w} \mathbf{B}_{ij}^j \mathbf{w}_j \right]}_{\text{bulk dynamics on } \Omega_i^c}
\end{aligned}$$

$$+ \underbrace{\left[\sum_{j \in \mathcal{N}_i^c} \left(\mathbf{K}_{ij}^i \mathbf{w}_i + \mathbf{K}_{ij}^j \mathbf{w}_j \right) \right]}_{\text{imperfect contact dynamics due to } \Gamma_r} + \underbrace{\mathbf{f}_i^{(c)}(t)}_{\text{heating BC}} \quad (59)$$

where $M = K + B$. For $k, l = 1, \dots, P$, the matrix elements are given by,

$$[\mathbf{A}_i]_{kl} = \int_{E_i} \rho_i c_{p,i} \phi_k^i \phi_l^i dE_i \quad (60a)$$

$$[\mathbf{B}_i]_{kl} = \int_{E_i} \mathbf{k}_i \nabla \phi_k^i \cdot \nabla \phi_l^i dE_i \quad (60b)$$

$$[\mathbf{B}_{ij}^i]_{kl} = \begin{cases} -\int_{e_{ij}} \{ \mathbf{k} \nabla \phi_k^i \cdot \mathbf{n}_i \} \phi_l^i + \sigma_b [\phi_k^i] \phi_l^i d\Gamma, & j \in \mathcal{N}_i^u \cup \{T\} \\ 0, & \text{otherwise} \end{cases} \quad (60c)$$

$$[\mathbf{B}_{ij}^j]_{kl} = \begin{cases} \int_{e_{ij}} \{ \mathbf{k} \nabla \phi_k^j \cdot \mathbf{n} \} \phi_l^j + \sigma_b [\phi_k^j] \phi_l^j d\Gamma, & j \in \mathcal{N}_i^u \cup \{T\} \\ 0, & \text{otherwise} \end{cases} \quad (60d)$$

$$[\mathbf{K}_{ij}^i]_{kl} = \begin{cases} -\int_{\Gamma_{ir}} \{ \mathbf{k} \nabla \phi_k^i \cdot \mathbf{n}_i \} \phi_l^i + \sigma_c [\phi_k^i] \phi_l^i d\Gamma, & j \in \mathcal{N}_i^c \\ 0, & \text{otherwise} \end{cases} \quad (60e)$$

$$[\mathbf{K}_{ij}^j]_{kl} = \begin{cases} \int_{\Gamma_{ir}} \{ \mathbf{k} \nabla \phi_k^j \cdot \mathbf{n} \} \phi_l^j + \sigma_c [\phi_k^j] \phi_l^j d\Gamma, & j \in \mathcal{N}_i^c \\ 0, & \text{otherwise} \end{cases} \quad (60f)$$

$$[\mathbf{f}_i^{(b)}]_k = \int_{e_{iq}} \phi_k^i q_b de + \sigma \int_{e_{iT}} \phi_k^i T_b de \quad (60g)$$

$$[\mathbf{f}_i^{(c)}]_k = \int_{\Gamma_{ir}} \phi_k^i q_c de + \sigma \int_{\Gamma_{iT}} \phi_k^i T_c de \quad (60h)$$

where the matrices $\mathbf{A}_i$, $\mathbf{B}_i$, $\mathbf{B}_{ij}^i$, and $\mathbf{B}_{ij}^j$ depend on ρ , c_p , and $\mathbf{k}$, respectively, which are non-linear functions of temperature and thus become the main source of non-linearities in the problem. The TCRs are assumed constant in this work, and thus the stiffness matrices due to imperfect contacts $\mathbf{K}_{ij}^i$ and $\mathbf{K}_{ij}^j$ are temperature-independent. Moreover, since the basis functions are orthogonal ρc_p is constant within an element, and $\mathbf{A}_i$ becomes a diagonal matrix; otherwise, $\mathbf{A}_i$ is symmetric and positive definite, as $\rho c_p > 0$.

For compactness, the element-wise ODEs in Eqs. (58) to (59) are stacked together to obtain the following system of ODEs for all bulk and interfacing elements in the TPS,

$$\mathcal{A}(\mathbf{x}) \dot{\mathbf{x}} = \mathcal{B}(\mathbf{x}) \mathbf{x} + \mathcal{F}(t) \quad (61)$$

where $\mathbf{x} = [\mathbf{u}, \mathbf{w}]^\top \in \mathbb{R}^{MP}$ are the states, and the matrices are given by,

$$\mathcal{A}(\mathbf{x}) = \begin{bmatrix} \mathbf{A}(\mathbf{u}) & \mathbf{0} \\ \mathbf{0} & \mathbf{D}(\mathbf{w}) \end{bmatrix}, \quad \mathcal{B}(\mathbf{x}) = \begin{bmatrix} \mathbf{B}(\mathbf{u}) & \mathbf{C}(\mathbf{u}) \\ \mathbf{E}(\mathbf{u}) & \mathbf{F}(\mathbf{w}) \end{bmatrix}, \quad \mathcal{F}(t) = \begin{bmatrix} \mathbf{f}^{(b)}(t) \\ \mathbf{f}^{(c)}(t) \end{bmatrix} \quad (62)$$

where $M = B + K$ and $\mathbf{B} \in \mathbb{R}^{BP \times BP}$, $\mathbf{C} \in \mathbb{R}^{BP \times KP}$, $\mathbf{E} \in \mathbb{R}^{KP \times BP}$, and $\mathbf{F} \in \mathbb{R}^{KP \times KP}$, with Neumann and Dirichlet boundary conditions assumed to be applied only to the bulk domains. Given that the neighboring elements $\mathcal{N}_i$ of E_i are sub-divided into bulk $\mathcal{N}_i^b = \mathcal{N}_i^u \cup \mathcal{N}_i^w$ and contact $\mathcal{N}_i^c$, the $M \times M$ block structure to the mass $\mathcal{A}$ and stiffness $\mathcal{B}$ matrices is introduced,

$$\mathbf{A}_{ij}(\mathbf{u}) = \begin{cases} \mathbf{A}_i(\mathbf{u}_i) & i = j, \\ \mathbf{0} & \text{otherwise} \end{cases}, \quad i, j = 1, \dots, B \quad (63a)$$

$$\mathbf{D}_{ij}(\mathbf{w}) = \begin{cases} \mathbf{D}_i(\mathbf{w}_i) & i = j, \\ \mathbf{0} & \text{otherwise} \end{cases}, \quad i, j = 1, \dots, K \quad (63b)$$

$$\mathbf{B}_{ij}(\mathbf{u}) = \begin{cases} \mathbf{B}_i(\mathbf{u}) + \sum_{l \in \mathcal{N}_i^b} \mathbf{B}_{il}^i(\mathbf{u}) & i = j, \\ \mathbf{B}_{ij}^j(\mathbf{u}) & i \neq j \text{ and } j \in \mathcal{N}_i^b \\ \mathbf{0} & \text{otherwise} \end{cases} \quad i, j = 1, \dots, B \quad (63c)$$

$$\mathbf{C}_{ij}(\mathbf{u}) = \begin{cases} \mathbf{B}_{ij}^j & j \in \mathcal{N}_i^w \\ \mathbf{0} & \text{otherwise} \end{cases} \quad i = 1, \dots, B, j = 1, \dots, K \quad (63d)$$

$$\mathbf{E}_{ij}(\mathbf{u}) = \begin{cases} \mathbf{B}_{ij}^j & j \in \mathcal{N}_i^u \\ \mathbf{0} & \text{otherwise} \end{cases} \quad i = 1, \dots, K, j = 1, 2, \dots, B \quad (63e)$$

$$\mathbf{F}_{ij}(\mathbf{w}) = \begin{cases} \mathbf{B}_i(\mathbf{w}_i) + \sum_{l \in \mathcal{N}_i^b} \mathbf{B}_{il}^i + \sum_{l \in \mathcal{N}_i^c} \mathbf{K}_{il}^i & i = j \\ \mathbf{B}_{ij}^j & i \neq j \text{ and } j \in \mathcal{N}_i^w \\ \mathbf{K}_{ij}^j & i \neq j \text{ and } j \in \mathcal{N}_i^c \\ \mathbf{0} & \text{otherwise} \end{cases} \quad i, j = 1, \dots, K \quad (63f)$$

A.2 Reduced-Physics Model

A *lumped capacitance model* (LCM) under imperfect TPS layer contacts is formulated in this section as the physics-based backbone of the PIROM formulation. The LCM is a classical physics-based low-order model for predicting the temporal variation of average temperatures in multiple interconnected components [INCROPERA]. The LCM is derived at the component level from a point of view of energy conservation, where the following assumptions are employed: (i) bulk Ω_i^b and interface Ω_i^c components have uniform temperatures $\bar{u}_i$, respectively, which are collected into an average variable $\bar{\mathbf{x}}_i = [\bar{u}_i, \bar{w}_i]^\top$; (ii) between neighboring i and j , the heat flux is given by $q_{ij} = g_{ij}(\bar{\mathbf{x}})(\bar{u}_i - \bar{u}_j)$ for $i, j \in \mathcal{N}_i^b$ through a *perfect*, and $q_{ij} = g_{x,ij}(\bar{w}_i - \bar{w}_j)$ for $i, j \in \mathcal{N}_i^c$ through an *imperfect* contact.

... in progress

A.3 Coarse-Graining Procedure

This section provides the details of coarse-graining for the TPS problem, the magnitude analysis of residual dynamics after coarse-graining, and an illustrative 2D example demonstrating the derivation of the LCM from the DG-FEM FOM.

A.3.1 Detailed Derivation of Coarse-Graining

This section supplements the details pertaining to the coarse-graining procedure presented in Sec. X. First, by definition,

$$\mathbf{r}^{(1)}(\bar{\mathbf{x}}, t) = \mathcal{P} \left[\mathbf{X}^\dagger \mathbf{r}(\mathbf{x}, t) \right] \quad (64a)$$

$$= \mathcal{P} \left[\mathbf{X}^\dagger \mathcal{A}(\mathbf{x})^{-1} \mathcal{B}(\mathbf{x}) \mathbf{x} + \mathbf{X}^\dagger \mathcal{A}(\mathbf{x})^{-1} \mathcal{F}(t) \right] \quad (64b)$$

$$= \mathcal{P} \left[\mathbf{X}^\dagger \mathcal{A}(\mathbf{x})^{-1} \mathbf{B}(\mathbf{x}) \mathbf{x} \right] + \mathcal{P} \left[\mathbf{X}^\dagger \mathcal{A}(\mathbf{x})^{-1} \mathcal{F}(t) \right] \quad (64c)$$

$$= \mathbf{X}^\dagger \mathcal{A}(\mathbf{P}\mathbf{x})^{-1} \mathbf{B}(\mathbf{P}\mathbf{x}) \mathbf{P}\mathbf{x} + \mathbf{X}^\dagger \mathcal{A}(\mathbf{P}\mathbf{x})^{-1} \mathbf{P}\mathcal{F}(t) \quad (64d)$$

$$= \underbrace{[\mathbf{X}^\dagger \mathbf{A}(\mathbf{X}\bar{\mathbf{x}})^{-1} \mathbf{X}]}_{\#1} \underbrace{[\mathbf{X}^\dagger \mathbf{B}(\mathbf{X}\bar{\mathbf{x}}) \mathbf{X}]}_{\#2} \bar{\mathbf{x}} + \mathbf{X}^\dagger \mathcal{A}(\mathbf{X}\bar{\mathbf{x}})^{-1} \mathbf{X} \underbrace{[\mathbf{X}^\dagger \mathcal{F}(t)]}_{\#3} \quad (64e)$$

where the terms #1, #2, and #3 are the coarse-grained representation of the inverted heat capacitance, heat conduction, and heat load matrices. Following the coarse-graining procedure developed in our previous work Ref. [VargasVenegas20206], these terms are treated one by one to demonstrate the equivalence between the LCM matrices and coarsened DG-FEM matrices.

Term #1: The capacitance matrix $\mathcal{A}$ is block diagonal, and composed of the bulk $\mathbf{A}$ and interface $\mathbf{D}$ capacitance matrices. These bulk and contact capacitance matrices are themselves diagonal, with their i -th diagonal blocks as $\mathbf{A}_i \in \mathbb{R}^{P \times P}$ and $\mathbf{D}_i \in \mathbb{R}^{P \times P}$, which become diagonal if the material properties are constant within the element. Take an element $E_i \in \Omega_j$ and denote $\overline{\rho c_{p_j}}$ as the value of ρc_p evaluated at the average $\bar{u}_j$ so that $\mathbf{A}_i = \overline{\rho c_{p_j}} |B_i| \mathbf{I}$ and $\mathbf{D}_i = \overline{\rho c_{p_j}} |W_i| \mathbf{I}$ with $|E_i|$ as the element size. The (k, l) -th element of the coarse-grained *bulk* and *contact* capacitance matrices are computed next.

Coarse-Grained Capacitance Matrices

The coarse-grained *bulk* capacitance matrix is computed as,

$$\left[\Phi^\dagger \mathbf{A} (\Phi \bar{\mathbf{u}})^{-1} \Phi \right]_{kl} = \sum_{i=1}^B \sum_{j=1}^B \varphi_i^{k\dagger} \mathbf{A}_{ij} (\Phi \bar{\mathbf{u}})^{-1} \varphi_j^l \quad (65a)$$

$$= \sum_{i=1}^B \sum_{j=1}^B \varphi_i^{k\dagger} \left[(\overline{\rho c_{p_k}} |B_i|)^{-1} \mathbf{I} \delta_{ij} \right] \varphi_j^l \quad (65b)$$

$$= \sum_{i=1}^B \varphi_i^{k\dagger} (\overline{\rho c_{p_k}} |B_i|)^{-1} \varphi_i^l \quad (65c)$$

$$= \sum_{i \in \mathcal{V}_k} \varphi_i^{k\dagger} (\overline{\rho c_{p_k}} |B_i|)^{-1} \varphi_i^l \quad (65d)$$

$$= \sum_{i \in \mathcal{V}_k} \frac{|B_i|}{|\Omega_k^b|} \varphi_i^{k\top} (\overline{\rho c_{p_k}} |B_i|)^{-1} \varphi_i^k \quad (65e)$$

$$= \sum_{i \in \mathcal{V}_k} \frac{1}{\Omega_k^b} \varphi_i^{k\top} (\overline{\rho c_{p_k}})^{-1} \varphi_i^k \quad (65f)$$

$$= \sum_{i \in \mathcal{V}_k} \left(\overline{\rho c_{p_k}} |\Omega_k^b| \right)^{-1} \quad (65g)$$

$$= |\mathcal{V}_k| \left(\overline{\rho c_{p_k}} |\Omega_k^b| \right)^{-1} \quad (65h)$$

Similarly, the coarse-grained *contact* capacitance matrix is computed as,

$$\left[\Pi^\dagger \mathbf{D} (\Pi \bar{\mathbf{w}})^{-1} \Pi \right]_{kl} = \sum_{i=1}^K \sum_{j=1}^K \psi_i^{k\dagger} \mathbf{D}_{ij} (\Pi \bar{\mathbf{w}})^{-1} \psi_j^l \quad (66a)$$

$$= \sum_{i=1}^K \sum_{j=1}^K \psi_i^{k\dagger} \left[(\overline{\rho c_{p_k}} |W_i|)^{-1} \mathbf{I} \delta_{ij} \right] \psi_j^l \quad (66b)$$

$$= \sum_{i=1}^K \psi_i^{k\dagger} (\overline{\rho c_{p_k}} |W_i|)^{-1} \psi_i^l \quad (66c)$$

$$= \sum_{i \in \mathcal{F}_k} \frac{|W_i|}{|\Omega_k^c|} \psi_i^{k\top} (\overline{\rho c_{p_k}} |W_i|)^{-1} \psi_i^k \quad (66d)$$

$$= \sum_{i \in \mathcal{F}_k} \frac{1}{|\Omega_k^c|} \psi_i^{k\top} (\overline{\rho c_{p_k}})^{-1} \psi_i^k \quad (66e)$$

$$= \sum_{i \in \mathcal{F}_k} (\overline{\rho c_{p_k}} |\Omega_k^c|)^{-1} \quad (66f)$$

$$= |\mathcal{F}_k| (\overline{\rho c_{p_k}} |\Omega_k^c|)^{-1} \quad (66g)$$

Therefore, by introducing the diagonal weight matrices $\mathbf{W}_V \in \mathbb{R}^{N_B \times N_B}$ and $\mathbf{W}_F \in \mathbb{R}^{N_C \times N_C}$ it follows that,

$$\Phi^\dagger \mathbf{A} (\Phi \bar{\mathbf{u}})^{-1} \Phi \mathbf{W}_V^{-1} = \bar{\mathbf{A}} (\bar{\mathbf{u}})^{-1}, \quad \Pi^\dagger \mathbf{D} (\Pi \bar{\mathbf{w}})^{-1} \Pi \mathbf{W}_F^{-1} = \bar{\mathbf{D}} (\bar{\mathbf{w}})^{-1} \quad (67)$$

Note that $|\Omega_k^c| \approx 0$ as the volume of the interfacing domains is reduced, and thus $\bar{\mathbf{D}} (\bar{\mathbf{w}}) \approx \mathbf{0}$ under a *zero-energy storage assumption for the interfacing domain*; such assumption is leveraged in subsequent derivations to demonstrate the equivalence between the coarse-grained DG-FEM model and Eq. (22).

Term #2: The thermal conductivity matrix $\mathbf{B}$ has a block structure that is divided into: $\mathbf{B} \in \mathbb{R}^{B \times B}$ capturing bulk heat transfer, $\mathbf{C} \in \mathbb{R}^{B \times K}$ and $\mathbf{E} \in \mathbb{R}^{K \times B}$ capturing the heat transfer coupling between bulk and contact sub-domains, and $\mathbf{F} \in \mathbb{R}^{K \times K}$ capturing the bulk heat transfer within the contact sub-domains. The (k, l) -th elements of the coarse-grained thermal conductivity matrices are computed next.

Coarse-Grained Thermal Conductivity Matrices

First, the stiffness matrices for the *bulk* and *contact* sub-domains are coarse-grained. The coarse-graining proceeds as follows,

$$\left[\Phi^\dagger \mathbf{B} (\Phi \bar{\mathbf{u}}) \Phi \right]_{kl} = \sum \quad (68a)$$

$$= \sum \quad (68b)$$

Similarly, coarse-graining the stiffness matrix for bulk heat transfer $\mathbf{F} (\mathbf{w})$ within the contact sub-domain proceeds as,

$$\left[\Pi^\dagger \mathbf{F} (\Pi \bar{\mathbf{w}}) \Pi \right]_{kl} = \sum_{i=1}^K \sum_{j=1}^K \psi_i^{k\top} \mathbf{F}_{ij} (\Pi \bar{\mathbf{w}}) \psi_j^l \quad (69a)$$

$$= \sum_{j \in \mathcal{F}_k} \sum_{j \in \mathcal{F}_l} \frac{|W_i|}{|\Omega_k^c|} \psi_i^{k\top} \mathbf{F}_{ij} (\Pi \bar{\mathbf{w}}) \psi_j^l \quad (69b)$$

$$= \frac{1}{|\mathcal{F}_k|} \sum_{i \in \mathcal{F}_k} \sum_{j \in \mathcal{F}_l} [\mathbf{F}_{ij}]_{11} \quad (69c)$$

Using the definition from Eq. (60h) it follows that,

$$[\mathbf{F}_{ij}]_{11} = \begin{cases} -\sum_{p \in \mathcal{N}_i^b} \sigma_b(\bar{\mathbf{u}}) |e_{ip}| - \sum_{p \in \mathcal{N}_i^c} \sigma_c |\Gamma_{ip}|, & i = j \\ \sigma_b(\bar{\mathbf{u}}) |e_{ij}|, & j \in \mathcal{N}_i^w \\ \sigma_c |\Gamma_{ij}|, & j \in \mathcal{N}_i^c \\ 0, & \text{otherwise} \end{cases} \quad (70)$$

Define the following sets collecting the edges and indices for interfaces with perfect contacts,

$$\mathcal{E}_{kl}^b = \{(i, j) \mid j \in \mathcal{N}_i^b, i \in \mathcal{F}_k\} \quad (71a)$$

$$\bar{\mathcal{N}}_k^b = \{l \mid \mathcal{E}_{kl}^b \neq \emptyset\} \quad (71b)$$

and imperfect contacts,

$$\mathcal{E}_{kl}^c = \{(i, j) \mid j \in \mathcal{N}_i^c, i \in \mathcal{F}_k, j \in \mathcal{F}_l\} \quad (72a)$$

$$\bar{\mathcal{N}}_k^c = \{l \mid \mathcal{E}_{kl}^c \neq \emptyset\} \quad (72b)$$

so that for $k, l = 1, 2, \dots, N_c$ the sum may be simplified as,

$$\sum_{i \in \mathcal{F}_k} \sum_{j \in \mathcal{F}_l} [\mathbf{F}_{ij}]_{11} = \begin{cases} -\sum_{p \in \bar{\mathcal{N}}_k^b} \sum_{(i,j) \in \mathcal{E}_{kp}^b} \sigma_b(\bar{\mathbf{u}}) |e_{ij}| - \sum_{p \in \bar{\mathcal{N}}_k^c} \sum_{(i,j) \in \mathcal{E}_{kp}^c} \sigma_c |\Gamma_{ij}|, & k = l \\ \sum_{(i,j) \in \mathcal{E}_{kl}^w} \sigma_b(\bar{\mathbf{u}}) |e_{ij}|, & l \in \bar{\mathcal{N}}_k^w \\ \sum_{(i,j) \in \mathcal{E}_{kl}^c} \sigma_c |\Gamma_{ij}|, & l \in \bar{\mathcal{N}}_k^c \\ 0, & \text{otherwise} \end{cases} \quad (73a)$$

$$= \begin{cases} -\sum_{p \in \bar{\mathcal{N}}_k^b} \sigma_b(\bar{\mathbf{u}}) |e_{kp}| - \sum_{p \in \bar{\mathcal{N}}_k^c} \sigma_c |\Gamma_{kp}|, & k = l \\ \sigma_b(\bar{\mathbf{u}}) |e_{kl}|, & l \in \bar{\mathcal{N}}_k^w \\ \sigma_c |\Gamma_{kl}|, & l \in \bar{\mathcal{N}}_k^c \\ 0, & \text{otherwise} \end{cases} \quad (73b)$$

Therefore, by introducing the diagonal weight matrix $\mathbf{W}_V \in \mathbb{R}^{N_c \times N_c}$, the coarse-grained stiffness matrix for the contact sub-domains is written as,

$$\bar{\mathbf{F}}(\bar{\mathbf{w}}) = \mathbf{W}_F \mathbf{\Pi}^\dagger \mathbf{F}(\mathbf{\Pi} \bar{\mathbf{w}}) \mathbf{\Pi} \quad (74)$$

The cross-coupling stiffness matrices $\mathbf{C} \in \mathbb{R}^{B \times K}$ and $\mathbf{E} \in \mathbb{R}^{K \times B}$ are coarse-grained next. The (k, l) -th element of the coarse-grained $\mathbf{C}$ matrix is computed as,

$$\left[\mathbf{\Phi}^\dagger \mathbf{C}(\mathbf{\Phi} \bar{\mathbf{u}}) \mathbf{\Pi} \right]_{kl} = \sum_{i=1}^B \sum_{j=1}^K \varphi_i^{k\dagger} \mathbf{C}_{ij}(\mathbf{\Phi} \bar{\mathbf{u}}) \psi_j^l \quad (75a)$$

$$= \sum_{i \in \mathcal{V}_k} \sum_{j \in \mathcal{F}_l} \varphi_i^{k\dagger} \mathbf{C}_{ij}(\mathbf{\Phi} \bar{\mathbf{u}}) \psi_j^l \quad (75b)$$

$$= \sum_{i \in \mathcal{V}_k} \sum_{j \in \mathcal{F}_l} \frac{|B_i|}{|\Omega_k^b|} \varphi_i^{k\top} \mathbf{C}_{ij}(\mathbf{\Phi} \bar{\mathbf{u}}) \psi_j^l \quad (75c)$$

$$= \sum_{i \in \mathcal{V}_k} \sum_{j \in \mathcal{F}_l} \frac{|B_i|}{|\Omega_k^b|} [\mathbf{C}_{ij}]_{11} \quad (75d)$$

$$= \frac{1}{|\mathcal{V}_k|} \sum_{i \in \mathcal{V}_k} \sum_{j \in \mathcal{F}_l} [\mathbf{B}_{ij}^j]_{11} \quad (75e)$$

Where it is assumed that all elements within a component have the same size, i.e., $|\Omega_k^b| = |B_i| |\mathcal{V}_k|$, the collection of edges between bulk Ω_k^b and contact Ω_l^c domains is $\mathcal{E}_{kl}^w = \{(i, j) | i \in \mathcal{V}_k, j \in \mathcal{N}_i^w, j \in \mathcal{F}_l\}$ so that $|e_{kl}| = \sum_{(i,j) \in \mathcal{E}_{kl}^w} |e_{ij}|$, and where the neighbors of component k are $\bar{\mathcal{N}}_k^w = \{l | \mathcal{E}_{kl}^w \neq \emptyset\}$. By leveraging the definition in Eq. (60h), the sum is simplified as,

$$\sum_{i \in \mathcal{V}_k} \sum_{j \in \mathcal{F}_l} [\mathbf{B}_{ij}^j]_{11} = \begin{cases} \sum_{(i,j) \in \mathcal{E}_{kl}^w} \sigma_b(\bar{\mathbf{u}}) |e_{ij}| = \sigma_b(\bar{\mathbf{u}}) |e_{kl}|, & l \in \bar{\mathcal{N}}_k^w \\ \mathbf{0}, & \text{otherwise} \end{cases} \quad (76)$$

Collecting for $k = 1, 2, \dots, N_B$ and $l = 1, 2, \dots, N_C$ leads to,

$$\bar{\mathbf{C}}(\bar{\mathbf{u}}) = \mathbf{W}_u \left(\Phi^\dagger \mathbf{C}(\Phi \bar{\mathbf{u}}) \Pi \right) \in \mathbb{R}^{N_B \times N_C} \quad (77)$$

with $\mathbf{W}_u$ as a diagonal weight matrix with $[\mathbf{W}_u]_i = |\mathcal{V}_k|$. Coarse-graining the $\mathbf{E}(\mathbf{u})$ matrix follows a similar procedure, where now $|\Omega_k^c| = |\mathcal{F}_k| |W_i|$, the neighbors on Ω_l^b of component k are $\bar{\mathcal{N}}_k^u = \{l | \mathcal{E}_{kl}^u \neq \emptyset\}$ with $\mathcal{E}_{kl}^u = \{(i, j) | i \in \mathcal{F}_k, j \in \mathcal{N}_i^u, j \in \mathcal{V}_l\}$, and therefore,

$$[\Pi^\dagger \mathbf{E}(\Phi \bar{\mathbf{u}}) \Phi]_{kl} = \sum_{i \in \mathcal{F}_k} \sum_{j \in \mathcal{V}_l} \frac{|W_i|}{|\Omega_k^c|} \psi_i^{k\top} \mathbf{E}_{ij}(\Phi \bar{\mathbf{u}}) \varphi_j^l \quad (78a)$$

$$= \sum_{i \in \mathcal{F}_k} \sum_{j \in \mathcal{V}_l} \frac{|W_i|}{|\Omega_k^c|} [\mathbf{E}_{ij}(\Phi \bar{\mathbf{u}})]_{11} \quad (78b)$$

$$= \frac{1}{|\mathcal{F}_k|} \sum_{i \in \mathcal{F}_k} \sum_{j \in \mathcal{V}_l} [\mathbf{B}_{ij}^j]_{11} \quad (78c)$$

From Eq. (76) it follows that,

$$\sum_{i \in \mathcal{F}_k} \sum_{j \in \mathcal{V}_l} [\mathbf{B}_{ij}^j]_{11} = \begin{cases} \sum_{p \in \bar{\mathcal{N}}_k^u} \sum_{(i,j) \in \mathcal{E}_{kp}^u} \sigma_b(\bar{\mathbf{u}}) |e_{ij}| = \sigma_b(\bar{\mathbf{u}}) |e_{kl}|, & l \in \bar{\mathcal{N}}_k^u \\ \mathbf{0}, & \text{otherwise} \end{cases} \quad (79)$$

Collecting for $k = 1, 2, \dots, N_C$ and $l = 1, 2, \dots, N_B$ leads to the following set of coarse-grained stiffness matrices,

$$\bar{\mathbf{B}} = \mathbf{W}_u \left(\Phi^\dagger \mathbf{B}(\Phi \bar{\mathbf{u}}) \Phi \right) \in \mathbb{R}^{N_B \times N_B} \quad (80a)$$

$$\bar{\mathbf{C}} = \mathbf{W}_u \left(\Phi^\dagger \mathbf{C}(\Phi \bar{\mathbf{u}}) \Pi \right) \in \mathbb{R}^{N_B \times N_C} \quad (80b)$$

$$\bar{\mathbf{E}} = \mathbf{W}_w \left(\Pi^\dagger \mathbf{E}(\Phi \bar{\mathbf{u}}) \Phi \right) \in \mathbb{R}^{N_C \times N_B} \quad (80c)$$

$$\bar{\mathbf{F}} = \mathbf{W}_w \left(\Pi^\dagger \mathbf{F}(\Pi \bar{\mathbf{w}}) \Pi \right) \in \mathbb{R}^{N_C \times N_C} \quad (80d)$$

Term #3: The heat-load vector $\mathcal{F}(t)$ is coarse-grained next. To simplify the derivation, it is assumed that the heat load is applied only to the bulk sub-domains, i.e., $\mathcal{F}(t) = [\mathbf{f}^{(b)}(t), \mathbf{0}]^\top$,

given that the contact sub-domains are assumed to be thin and have negligible energy storage in subsequent developments. Therefore, the k -th element of the coarse-grained heat-load vector is computed as,

Coarse-Grained Heat-Load Vector

$$\left[\Phi^\dagger \mathcal{F}(t) \right]_k = \sum_{i=1}^B \varphi_i^{k\dagger} \mathbf{f}_i^{(b)}(t) \quad (81a)$$

$$= \sum_{i \in \mathcal{V}_k} \frac{|B_i|}{|\Omega_k^b|} \varphi_i^{k\dagger} \mathbf{f}_i^{(b)} \quad (81b)$$

$$= \frac{1}{|\mathcal{V}_k|} \sum_{i \in \mathcal{V}_k} \left[\mathbf{f}_i^{(b)} \right]_1 \quad (81c)$$

$$= \frac{1}{|\mathcal{V}_k|} \sum_{i \in \mathcal{V}_k} |e_{iq}| \bar{q}_{i,b}(t) \quad (81d)$$

$$= \frac{1}{|\mathcal{V}_k|} |e_{kq}| \bar{q}_{k,b}(t) \quad (81e)$$

Collecting for $k = 1, 2, \dots, N_B$ leads to the following coarse-grained heat-load vector,

$$\bar{\mathbf{F}}(t) = \mathbf{W}_u \left(\Phi^\dagger \mathcal{F}(t) \right) \in \mathbb{R}^{N_B} \quad (82)$$

By collecting the coarse-grained capacitance and stiffness matrices, and applying the zero-energy storage assumption, the following DAE system is obtained,

$$\bar{\mathbf{A}}(\bar{\mathbf{u}}) \dot{\bar{\mathbf{u}}} = \bar{\mathbf{B}}\bar{\mathbf{u}} + \bar{\mathbf{C}}\bar{\mathbf{w}} + \bar{\mathbf{F}}(t) \quad (83a)$$

$$\mathbf{0} = \bar{\mathbf{E}}(\bar{\mathbf{u}}) \bar{\mathbf{u}} + \bar{\mathbf{F}}\bar{\mathbf{w}} \quad (83b)$$

where Eq. (83b) may be re-arranged to match precisely the energy conservation from Eq. (15). Specifically, the k -th equation of Eq. (83b) states that,

$$0 = \sum_{p \in \tilde{\mathcal{N}}_k^u} \sigma_b(\bar{\mathbf{u}}) |e_{kp}| (\bar{u}_p - \bar{w}_k) - \sum_{p \in \tilde{\mathcal{N}}_k^c} \sigma_c |\Gamma_{kp}| (\bar{w}_k - \bar{w}_p) \quad (84)$$

Now, note that each domain is decomposed as $\Omega_k = \Omega_k^b \cup \Omega_k^c$, and that Γ_r for $r = 1, 2, \dots, N_\Gamma$ is composed of two unique interface domains $\Omega_r^{c,-} \cup \Omega_r^{c,+}$. Therefore, Eq. (84) simplifies to,

$$0 = \sigma_b(\bar{\mathbf{u}}) |e_{kk}| (\bar{u}_k - \bar{w}_k) - \sigma_{c,r} |\Gamma_r| \Delta \bar{w}_r \quad (85)$$

which corresponds precisely to the heat flux continuity constraint in Eq. (15), from which Eq. (16) is derived. The derivation of Eq. (22) from the coarse-grained DG-FEM model is further demonstrated next in the following example.

Coarse-Graining Example Consider the three-layered TPS shown in Fig. [X] with three components $\{\Omega_j\}_{j=1}^3$ and three contact interfaces $\{\Gamma_k\}_{k=1}^3$. The set of elements for each component, and on the minus and plus sides of each interface are defined as,

$$\mathcal{V}_j = \{m \in \{1, \dots, M\} : E_m \subseteq \Omega_j\}, \quad (86a)$$

$$\mathcal{F}_k^- = \{m \in \{1, \dots, M\} : E_m \cap \Gamma_k \neq \emptyset, E_m \subseteq \Omega_{l(k)}\}, \quad (86b)$$

$$\mathcal{F}_k^+ = \{m \in \{1, \dots, M\} : E_m \cap \Gamma_k \neq \emptyset, E_m \subseteq \Omega_{r(k)}\} \quad (86c)$$

where $\Omega_{l(k)}$ and $\Omega_{r(k)}$ are the left and right components that form the interface Γ_k , respectively, with $\Gamma_k = \partial\Omega_{l(k)} \cap \partial\Omega_{r(k)}$.

Important point: The critical assumption is that the contact domains $\Omega_r^{c,-}$ and $\Omega_r^{c,+}$ are sufficiently thin such that their transient thermal response is negligible, and thus the contact temperature jumps $\Delta\bar{\theta}_r$ are governed by algebraic constraints for heat-flux continuity across the interfaces.